(ST 599) Missing Data & Casual Inference
Final Project

Introduction

Missing data is a arguably always present and often a challenge in statistical analysis. In the solving of such a problem, imputation is a common strategy, then to choose a correct model, selection techniques are often employed. This project investigates the impact of combining resampling methods with various imputation techniques and their effects on model selection under different missingness mechanisms: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). A central goal of this study is to assess the accuracy of model selection, determine whether resampling improves model accuracy, and compare the biases introduced by missing data under different imputation and model selection strategies. Specifically, the project explores how often the correct model is selected under different combinations of resampling, imputation, and model selection techniques, and whether these strategies help mitigate the biases caused by missing data.

By evaluating the performance of various imputation methods, resampling techniques, and model selection procedures, this study aims to compare and contrast multiple factors used in model selection and general statistical analysis.

Methodology

Simulation Study Design

The simulation study is designed to assess the impact of resampling, imputation, and model selection methods under different missingness mechanisms. The different levels with a brief description for each factor under evaluation are presented below

- **Missingness Mechanisms**
  - **MCAR (Missing Completely at Random):** Data points are removed randomly with a fixed probability, meaning missingness is independent of both observed and unobserved data.
  - **MNAR (Missing Not at Random):** The probability of missingness depends on the value of the missing data itself, even after accounting for observed variables. This introduces bias because the missing data mechanism cannot be fully accounted for using available information.
  - **MAR (Missing at Random):** Missingness depends on observed variables but not on the missing values themselves. That is, after conditioning on observed data, the probability of missingness is unrelated to the unobserved values.
    - See R-Code File 1 in the Appendix for specifics on the missingness mechanisms
- **Resampling Techniques**
  - **Bootstrap:** A resampling technique that repeatedly draws B samples with replacement from the original dataset.
  - **Jackknife:** A resampling method that systematically leaves out one observation at a time, in our case leading to N resampled datasets.

    - Due to computational feasibility and final model counts climbing due to possible combinations, jackknife is not explored in this project.

- **Imputation Procedures**
  - **Deterministic Mean Imputation:** Missing values are replaced with the mean of the observed values for that variable.
  - **Stochastic Mean Imputation:** Similar to mean imputation above but adds random noise, $\text{Normal}\,(0,\ \text{SE}\,(X_i))$, from the observed data distribution.
  - **Deterministic K-Nearest Neighbors (KNN) Imputation:** Missing values are replaced using the average of the k nearest observed neighbors based on a distance metric (Euclidean distance).
    - An arbitrary value for k was chosen to be 3 throughout these simulations
  - **Stochastic "K-Nearest Neighbors (KNN)" Imputation:** Instead of taking the average, among the k nearest neighbors, one is chosen at random.
  - **Deterministic Regression Imputation:** Missing values are imputed using predicted values from a regression model trained on observed data.
  - **Stochastic Regression Imputation:** Similar to deterministic regression imputation but includes a randomly drawn residual term based on the fitted model's standard error.

- **Model Selection Procedures**
  - **LASSO:** A regularization method that applies an L_1-penalty to shrink some coefficients to zero, performing variable selection.
    - Cross validation was performed to select the tuning parameter.
  - **Ridge:** A regularization method that applies an L_2-penalty, shrinking coefficients continuously without setting any to zero, thus does not perform variable selection.
    - Cross validation was performed to select the tuning parameter.
  - **Forward Selection:** A stepwise selection approach where variables are added one at a time based on a specified information criterion (AIC / BIC) until no further improvement can be achieved.
  - **Best Subsets:** Evaluates all possible combinations of predictors and selects the model that optimizes some predefined criterion (AIC / BIC).

- **Model Selection Resampling Combinations Techniques**
  - **Stability Selection:** A method that applies resampling and selection procedures multiple times to identify variables that consistently appear in selected models.
    - An arbitrary value of 0.50 was used as a cutoff of whether to include the variable in the model.
    - In this project's context the fitted model based on stability selection uses the original data (not resampled).
    - Also note that LASSO and Ridge are not used in combination with stability selection.

- **Stacking:** The combination of all the resampled datasets into one for the model selection procedure.

The following outlines a leave-one-out methodological procedure to try and parse the effect of each component of the full method.

(1) **Data Generation:** Simulated datasets are created where the covariates follow some multivariate normal distribution, and the response variable is generated using a linear model with predefined regression coefficients.

(2) **Inducing Missingness:** Missing values are introduced under one of the three different missingness mechanisms for both the response and covariates.

(3) **Scenario 1: Resampling and Model Selection without Imputation**

   (a) **Resampling Procedures:** Bootstrap is applied to the generated data.

   (b) **Complete Cases:** Save only the complete cases for each resampled datasets.

   (c) **Model Selection:** Model selection techniques described above are performed on each resampled dataset (for both resampling methods).

   (d) **Model Combination:** Stability selection and stacking are performed across resampled datasets for resulting models.

   (e) **Final Models:** 18 final models are then obtained (71 under all possible simulation settings)

(4) **Scenario 2: Model Selection with Imputation but Without Resampling**

   (a) **Imputation Techniques:** Missing values are imputed using each of the imputation techniques outlined above.

   (b) **Model Selection:** Model selection techniques described above are performed on each imputed dataset.

   (c) **Final Models:** 72 final models are then obtained (288 under all possible simulation settings)

(5) **Scenario 3: Model Selection without Resampling or Imputation**

   (a) **Complete Cases:** Complete cases of the generated dataset.

   (b) **Model Selection:** Perform model selection techniques

   (c) **Final Models:** 12 final models are then obtained (48 under all possible simulation settings)

(6) **Scenario 4: Model Selection with Both Resampling and Imputation**

   (a) **Resampling Procedures:** The generated data is bootstrapped.

   (b) **Imputation:** Each resampled dataset is imputed using each of the imputation techniques.

   (c) **Model Selection:** Model selection techniques are performed.

(d) **Model Combination:** Stability selection and stacking are performed across resampled datasets for resulting models.

(e) **Final Models:** 108 final models are then obtained (432 under all possible simulation settings)

The above steps are repeated many times for the different simulation settings described below and results are collected. Results include model summary outputs and selection frequencies.

Simulation Settings

Due to the already very large number of final models only a few simulation settings are chosen in which covariance and underlying missingness is manipulated. For all simulations there 5 covariates variables $\boldsymbol{X}$ that follow a multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ which depends on the correlation setting. For the independent settings we have

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \boldsymbol{0}_5,\, \boldsymbol{\Sigma}_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}\right), \quad i, j = 1, \ldots 5$$

$$\boldsymbol{\beta}^T = \begin{pmatrix} 0 & 0 & 0.5 & 1 & 1.5 & 2 \end{pmatrix}, \quad \boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta}, \quad \boldsymbol{\epsilon} \sim N(0, 1)$$

and for the moderate correlation settings we have

$$\boldsymbol{X} \sim \boldsymbol{MVN}\left(\boldsymbol{\mu} = \boldsymbol{0}_5,\, \boldsymbol{\Sigma}_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0.666 & \text{if } i \neq j. \end{cases}\right), \quad i, j = 1, \ldots 5$$

$$\boldsymbol{\beta}^T = \begin{pmatrix} 0 & 0 & 0.5 & 1 & 1.5 & 2 \end{pmatrix}, \quad \boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta}, \quad \boldsymbol{\epsilon} \sim N(0, 1)$$

The missingness proportions are categorized as low and high where low is 10% missing and high is 25% missing. The different simulation settings are summarized in Table 1 below and specifics on the data generating process can be found in the R-Code File 1 in the Appendix.

Covariate Correlation	**Missing Proportion**	
	Low	**High**
Independent	Setting 1	Setting 2
Moderately Correlated	Setting 3	Setting 4

Table 1: Combinations of Covariate Correlation and Missingness Levels

Results and Discussion

Scenario 1

The results in Tables 2 and 3 present the final models for Scenario 1, each employing a different combination of variable selection, stability selection, and stacking. Notably, stacking results in a

significantly higher frequency of β_1 in the model, despite being simulated as zero. In fact, stacking tends to include all variables regardless of the setting. However, the average estimated coefficient values remain close to the true values across both tables.

Scenario 2

Tables 4 through 9 summarize the final models for Scenario 2, each corresponding to a different imputation method. Both deterministic and stochastic mean and KNN imputation perform poorly, deviating substantially from the true coefficient values. In contrast, regression imputation—both deterministic and stochastic—performs well under all missingness mechanisms. No substantial differences were observed between deterministic and stochastic imputation within the examined simulation settings.

Scenario 3

Table 10 presents the results for Scenario 3, where no imputation or resampling was performed. Interestingly, the average estimated coefficients remain close to the true values, and the variable selection frequencies appear reasonable, making this outcome somewhat unexpected.

Scenario 4

Tables 11 through 22 display results for Scenario 4, which involves both imputation and resampling. This scenario encompasses 12 tables, each representing a different imputation method combined with either stability selection or stacking.

- In Table 11, deterministic mean imputation with stability selection produces accurate coefficient estimates under all missingness mechanisms. However, Table 12 reveals that pairing deterministic mean imputation with stacking leads to downward-biased coefficients, particularly under MAR.
- A similar pattern emerges in Tables 13 and 14 (stochastic mean imputation with stability selection and stacking, respectively), where stability selection yields reliable coefficient estimates, but stacking introduces bias. Notably, Table 14 shows less differentiation across missingness mechanisms.
- Tables 15 through 18 (deterministic and stochastic KNN imputation using stability selection and stacking) further reinforce the recurring pattern of stability selection performing well, while stacking increases the inclusion of unrelated variables.
- Finally, Tables 19 through 22 (deterministic and stochastic regression imputation with stability selection and stacking) indicate that regression imputation reduces differences between stability and stacking. However, stacking still leads to the inclusion of more unrelated covariates.

Comparison Across Scenarios

Scenarios without resampling show that Forward and Best Subsets selection produce identical final models, leading to the same coefficient estimates and variable selection frequencies. Regression imputation—both deterministic and stochastic—outperforms other imputation methods in coefficient

estimation and variable selection. Interestingly, the scenario with no imputation or resampling yields surprisingly accurate coefficient estimates, warranting further investigation.

Conclusion

This project has explored the impact of imputation methods, resampling techniques, and model selection procedures under various missingness mechanisms (MCAR, MAR, and MNAR). Overall, the results suggest that regression imputation might be superior to the other imputation methods tested by improving the accuracy of model selection and coefficient estimation. There is also evidence to suggest that stability selection appears to perform better than stacking, as the latter tends to include irrelevant variables, potentially reducing model accuracy. Further research is needed to refine model selection techniques and investigate the effects of the missingness mechanisms as well as the deterministic versus stochastic imputation in the model selection context.

Appendix

Table Notation

- Column Notation

Notation	Full Name
Miss Mech	Missingness Mechanism (MCAR, MNAR, MAR)
Miss Rate	Missing Rate / Proportion (Low, High)
Cor	Correlation (Ind, Corr)
Sel	Model Selection Method
$\hat{\beta}$ NA 0 %	For the given $\hat{\beta}$ coefficient estimate, (NA) is the average calculated by removing NAs, (0) is the average calculated by treated NAs as 0s, (%) is the proportion of times the coefficient was chosen

- Row Notation

Notation	Full Name
MCAR MNAR MAR	Missing Completely At Random Missing Not At Random Missing At Random
Low High	Low Missingness Proportion (10%) High Missingness Proportion (25%)
Ind Corr	Independence Between Covariates Moderate Correlation ($\approx$ 66%)
LASSO Ridge Forward BestSub	Least Absolute Shrinkage and Selection Operator Ridge Regression Forward Stepwise Selection Best Subset Selection

- Tables 2 to 22 follow a similar structure of displaying the average estimated coefficients for the different simulation settings and scenarios.

Table 2: **Scenario 1 (No Imputation) | Combination Method - Stability Selection**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.01	-0.00	18	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.05	0.01	19	0.51	0.49	96	1.00	0.99	100	1.50	1.50	100	2.01	2.01	100
			BestSub	0.00	0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.03	-0.01	24	0.52	0.48	93	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.06	0.01	22	0.60	0.44	73	1.01	0.99	98	1.52	1.52	100	2.02	2.02	100
			BestSub	-0.00	-0.00	100	0.01	0.01	100	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	-0.00	-0.00	17	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	0.01	0.00	20	0.51	0.49	97	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.02	-0.00	20	0.52	0.48	93	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.04	-0.04	100	0.07	0.02	22	0.58	0.46	79	1.00	1.00	100	1.51	1.51	100	2.00	2.00	100
			BestSub	-0.04	-0.04	100	0.01	0.01	100	0.50	0.50	100	0.99	0.98	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.02	-0.00	19	0.51	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	-0.00	-0.00	100	0.01	0.00	20	0.51	0.50	97	1.00	1.00	100	1.51	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.02	-0.00	20	0.52	0.49	94	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.01	-0.01	100	0.02	0.00	21	0.55	0.46	84	1.00	1.00	100	1.51	1.51	100	2.01	2.01	100
			BestSub	-0.00	-0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100

Table 3: **Scenario 1 (No Imputation) | Combination Method - Stacking**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.49	0.49		1.00	1.00		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.00	-0.00		0.47	0.47		0.94	0.94		1.40	1.40		1.87	1.86	
			Forward	0.00	0.00	100	-0.00	-0.00	90	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	90	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.03	0.03		0.48	0.48		0.98	0.98		1.49	1.49		1.99	1.99	
			Ridge		0.00		0.20	0.20		0.58	0.58		0.97	0.97		1.37	1.37		1.76	1.76	
			Forward	0.00	0.00	100	0.00	0.00	89	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.00	100
			BestSub	0.00	0.00	100	0.00	0.00	89	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.00	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.49	0.49		0.99	0.99		1.50	1.49		1.98	1.98	
			Ridge		0.00		-0.01	-0.01		0.46	0.46		0.93	0.93		1.40	1.39		1.84	1.84	
			Forward	-0.00	-0.00	100	-0.01	-0.01	89	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	89	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**LASSO**		0.00		0.03	0.03		0.50	0.50		0.97	0.97		1.50	1.49		1.99	1.99	
			Ridge		0.00		0.22	0.22		0.60	0.60		0.96	0.96		1.37	1.37		1.74	1.74	
			Forward	-0.00	-0.00	100	0.01	0.01	90	0.52	0.51	98	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
			BestSub	-0.00	-0.00	100	0.01	0.01	90	0.52	0.51	98	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.50	0.49		1.00	0.99		1.49	1.49		1.99	1.98	
			Ridge		0.00		-0.00	-0.00		0.47	0.47		0.94	0.94		1.40	1.40		1.86	1.86	
			Forward	-0.02	-0.02	100	-0.01	-0.00	89	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.00	89	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**LASSO**		0.00		0.02	0.02		0.49	0.49		0.98	0.98		1.49	1.49		1.99	1.99	
			Ridge		0.00		0.19	0.19		0.58	0.58		0.97	0.97		1.37	1.37		1.77	1.76	
			Forward	-0.01	-0.01	100	-0.00	-0.00	90	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	90	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.49	0.49		0.99	0.99		1.49	1.48		1.97	1.97	
			Ridge		0.00		-0.01	-0.01		0.47	0.47		0.92	0.92		1.39	1.39		1.84	1.84	
			Forward	-0.07	-0.07	100	-0.01	-0.01	89	0.51	0.50	99	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	89	0.51	0.50	99	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**LASSO**		0.00		0.03	0.03		0.49	0.49		0.98	0.97		1.49	1.49		1.98	1.98	
			Ridge		0.00		0.17	0.17		0.57	0.57		0.96	0.96		1.38	1.38		1.78	1.78	
			Forward	-0.04	-0.04	100	0.01	0.01	88	0.51	0.50	99	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.04	-0.04	100	0.01	0.01	88	0.51	0.50	99	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.50	0.50		0.99	0.99		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.00	-0.00		0.47	0.47		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	0.00	0.00	100	-0.00	-0.00	90	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	90	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.02	0.02		0.49	0.49		0.98	0.98		1.50	1.49		1.99	1.99	
			Ridge		0.00		0.17	0.17		0.58	0.58		0.97	0.97		1.39	1.38		1.78	1.78	
			Forward	0.00	0.00	100	-0.00	-0.00	89	0.51	0.50	100	1.00	1.00	100	1.51	1.51	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	89	0.51	0.50	100	1.00	1.00	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.49	0.49		0.99	0.99		1.49	1.48		2.00	1.99	
			Ridge		0.00		-0.01	-0.01		0.46	0.46		0.92	0.92		1.39	1.38		1.86	1.86	
			Forward	-0.01	-0.01	100	-0.01	-0.01	90	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	90	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.01	0.01		0.49	0.49		0.98	0.98		1.49	1.49		2.00	2.00	
			Ridge		0.00		0.12	0.12		0.53	0.53		0.97	0.96		1.41	1.41		1.82	1.82	
			Forward	-0.00	-0.00	100	-0.00	-0.00	88	0.50	0.50	99	0.99	0.99	100	1.50	1.50	100	2.01	2.00	100
			BestSub	-0.00	-0.00	100	-0.00	-0.00	88	0.50	0.50	99	0.99	0.99	100	1.50	1.50	100	2.01	2.00	100

Table 4: **Scenario 2 (No Resampling) | Imputation Method - Deterministic Mean Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.43	0.43		0.88	0.88		1.33	1.33		1.78	1.78	
			Ridge		0.00		-0.01	-0.01		0.43	0.43		0.85	0.85		1.28	1.27		1.70	1.69	
			Forward	0.00	0.00	100	-0.02	-0.00	16	0.46	0.45	99	0.91	0.90	100	1.35	1.35	100	1.80	1.80	100
			BestSub	0.00	0.00	100	-0.02	-0.00	16	0.46	0.45	99	0.91	0.90	100	1.35	1.35	100	1.80	1.80	100
		Corr	**LASSO**		0.00		0.23	0.23		0.58	0.58		0.96	0.96		1.33	1.33		1.71	1.71	
			Ridge		0.00		0.33	0.33		0.64	0.64		0.95	0.95		1.25	1.25		1.56	1.56	
			Forward	-0.00	-0.00	100	0.42	0.17	41	0.64	0.59	92	0.99	0.99	100	1.35	1.35	100	1.73	1.73	100
			BestSub	-0.00	-0.00	100	0.42	0.17	41	0.64	0.59	92	0.99	0.99	100	1.35	1.35	100	1.73	1.73	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.35	0.35		0.72	0.72		1.10	1.09		1.47	1.47	
			Ridge		0.00		-0.01	-0.01		0.36	0.36		0.71	0.71		1.07	1.07		1.42	1.42	
			Forward	0.00	0.00	100	-0.03	-0.01	19	0.41	0.36	87	0.75	0.75	100	1.12	1.12	100	1.50	1.50	100
			BestSub	0.00	0.00	100	-0.03	-0.01	19	0.41	0.36	87	0.75	0.75	100	1.12	1.12	100	1.50	1.50	100
		Corr	**LASSO**		0.00		0.43	0.43		0.65	0.65		0.90	0.90		1.13	1.13		1.41	1.41	
			Ridge		0.00		0.49	0.49		0.67	0.67		0.87	0.87		1.06	1.06		1.27	1.27	
			Forward	-0.01	-0.01	100	0.62	0.38	61	0.76	0.64	85	0.96	0.93	97	1.17	1.17	100	1.44	1.44	100
			BestSub	-0.01	-0.01	100	0.62	0.38	61	0.76	0.64	85	0.96	0.93	97	1.17	1.17	100	1.44	1.44	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.43	0.43		0.88	0.88		1.33	1.32		1.77	1.77	
			Ridge		0.00		-0.00	-0.00		0.42	0.42		0.85	0.84		1.27	1.27		1.69	1.69	
			Forward	0.10	0.10	100	0.00	0.00	14	0.45	0.45	99	0.90	0.90	100	1.35	1.35	100	1.79	1.79	100
			BestSub	0.10	0.10	100	0.00	0.00	14	0.45	0.45	99	0.90	0.90	100	1.35	1.35	100	1.79	1.79	100
		Corr	**LASSO**		0.00		0.22	0.22		0.58	0.58		0.96	0.96		1.34	1.34		1.72	1.72	
			Ridge		0.00		0.33	0.33		0.63	0.63		0.95	0.95		1.26	1.26		1.57	1.57	
			Forward	0.03	0.03	100	0.42	0.17	41	0.63	0.59	93	0.99	0.99	100	1.36	1.36	100	1.75	1.74	100
			BestSub	0.03	0.03	100	0.42	0.17	41	0.63	0.59	93	0.99	0.99	100	1.36	1.36	100	1.75	1.74	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.34	0.34		0.71	0.71		1.08	1.08		1.46	1.46	
			Ridge		0.00		-0.00	-0.00		0.35	0.35		0.70	0.70		1.05	1.05		1.41	1.41	
			Forward	0.15	0.15	100	-0.01	-0.00	16	0.41	0.35	85	0.74	0.74	100	1.11	1.11	100	1.49	1.49	100
			BestSub	0.15	0.15	100	-0.01	-0.00	16	0.41	0.35	85	0.74	0.74	100	1.11	1.11	100	1.49	1.49	100
		Corr	**LASSO**		0.00		0.42	0.42		0.64	0.64		0.90	0.90		1.18	1.18		1.44	1.44	
			Ridge		0.00		0.48	0.48		0.67	0.67		0.87	0.87		1.10	1.10		1.30	1.30	
			Forward	0.04	0.04	100	0.62	0.37	60	0.75	0.64	85	0.95	0.93	97	1.22	1.21	99	1.47	1.47	100
			BestSub	0.04	0.04	100	0.62	0.37	60	0.75	0.64	85	0.95	0.93	97	1.22	1.21	99	1.47	1.47	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.39	0.39		0.78	0.78		1.17	1.17		1.58	1.58	
			Ridge		0.00		-0.00	-0.00		0.39	0.39		0.75	0.75		1.12	1.11		1.50	1.50	
			Forward	0.00	0.00	100	0.01	0.00	15	0.42	0.41	98	0.80	0.80	100	1.19	1.18	100	1.60	1.59	100
			BestSub	0.00	0.00	100	0.01	0.00	15	0.42	0.41	98	0.80	0.80	100	1.19	1.18	100	1.60	1.59	100
		Corr	**LASSO**		0.00		0.17	0.17		0.38	0.38		0.72	0.72		1.13	1.13		1.55	1.55	
			Ridge		0.00		0.24	0.24		0.45	0.45		0.76	0.76		1.08	1.08		1.41	1.41	
			Forward	-0.64	-0.64	100	0.38	0.13	34	0.55	0.35	64	0.78	0.75	96	1.17	1.16	100	1.59	1.59	100
			BestSub	-0.64	-0.64	100	0.38	0.13	34	0.55	0.35	64	0.78	0.75	96	1.17	1.16	100	1.59	1.59	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.31	0.31		0.41	0.41		0.63	0.63		0.97	0.96	
			Ridge		0.00		-0.01	-0.01		0.33	0.33		0.41	0.41		0.62	0.62		0.94	0.94	
			Forward	-0.03	-0.03	100	-0.02	-0.00	16	0.43	0.31	73	0.44	0.43	97	0.65	0.65	100	0.99	0.99	100
			BestSub	-0.03	-0.03	100	-0.02	-0.00	16	0.43	0.31	73	0.44	0.43	97	0.65	0.65	100	0.99	0.99	100
		Corr	**LASSO**		0.00		0.85	0.85		0.99	0.99		0.04	0.04		0.25	0.25		0.72	0.72	
			Ridge		0.00		0.82	0.82		0.94	0.94		0.08	0.08		0.28	0.28		0.64	0.64	
			Forward	-1.72	-1.72	100	0.91	0.88	97	1.04	1.03	99	0.12	0.02	17	0.41	0.20	49	0.79	0.78	99
			BestSub	-1.72	-1.72	100	0.91	0.88	97	1.04	1.03	99	0.12	0.02	17	0.41	0.20	49	0.79	0.78	99

Table 5: **Scenario 2 (No Resampling) | Imputation Method - Stochastic Mean Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.38	0.38		0.77	0.76		1.16	1.16		1.58	1.57	
			Ridge		0.00		-0.00	-0.00		0.39	0.39		0.76	0.76		1.14	1.14		1.53	1.52	
			Forward	0.00	0.00	100	-0.03	-0.01	17	0.53	0.36	68	0.83	0.81	97	1.21	1.21	100	1.63	1.62	100
			BestSub	0.00	0.00	100	-0.03	-0.01	17	0.53	0.36	68	0.83	0.81	97	1.21	1.21	100	1.63	1.62	100
		Corr	**LASSO**		0.00		0.41	0.41		0.63	0.63		0.92	0.92		1.15	1.15		1.41	1.41	
			Ridge		0.00		0.48	0.48		0.66	0.66		0.88	0.87		1.02	1.02		1.19	1.19	
			Forward	-0.00	-0.00	100	1.19	0.34	28	1.32	0.57	43	1.44	0.93	64	1.61	1.21	75	1.80	1.53	85
			BestSub	-0.00	-0.00	100	1.19	0.34	28	1.32	0.57	43	1.44	0.93	64	1.61	1.21	75	1.80	1.53	85
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.21	0.21		0.46	0.46		0.69	0.69		1.00	1.00	
			Ridge		0.00		-0.00	-0.00		0.21	0.21		0.45	0.45		0.64	0.64		0.90	0.89	
			Forward	-0.00	-0.00	100	-0.02	-0.00	18	0.61	0.18	30	0.79	0.48	62	0.93	0.78	84	1.19	1.13	95
			BestSub	-0.00	-0.00	100	-0.02	-0.00	18	0.61	0.18	30	0.79	0.48	62	0.93	0.78	84	1.19	1.13	95
		Corr	**LASSO**		0.00		0.46	0.46		0.53	0.53		0.66	0.66		0.79	0.79		0.93	0.93	
			Ridge		0.00		0.47	0.47		0.51	0.51		0.61	0.61		0.71	0.71		0.78	0.78	
			Forward	-0.03	-0.03	100	1.67	0.46	28	1.70	0.57	33	1.76	0.73	41	1.88	0.87	46	1.93	1.07	55
			BestSub	-0.03	-0.03	100	1.67	0.46	28	1.70	0.57	33	1.76	0.73	41	1.88	0.87	46	1.93	1.07	55
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.37	0.37		0.77	0.77		1.17	1.16		1.57	1.57	
			Ridge		0.00		-0.01	-0.01		0.38	0.38		0.77	0.77		1.14	1.14		1.52	1.52	
			Forward	0.07	0.07	100	-0.03	-0.00	17	0.53	0.35	67	0.84	0.81	97	1.22	1.21	100	1.62	1.62	100
			BestSub	0.07	0.07	100	-0.03	-0.00	17	0.53	0.35	67	0.84	0.81	97	1.22	1.21	100	1.62	1.62	100
		Corr	**LASSO**		0.00		0.39	0.39		0.64	0.64		0.87	0.87		1.16	1.16		1.44	1.44	
			Ridge		0.00		0.49	0.49		0.66	0.66		0.84	0.84		1.03	1.03		1.21	1.21	
			Forward	0.00	0.00	100	1.12	0.32	29	1.31	0.60	46	1.41	0.86	61	1.62	1.23	76	1.84	1.58	86
			BestSub	0.00	0.00	100	1.12	0.32	29	1.31	0.60	46	1.41	0.86	61	1.62	1.23	76	1.84	1.58	86
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.21	0.21		0.46	0.46		0.68	0.68		0.98	0.98	
			Ridge		0.00		-0.01	-0.01		0.21	0.21		0.44	0.44		0.64	0.64		0.87	0.87	
			Forward	-0.07	-0.07	100	-0.11	-0.02	16	0.58	0.17	30	0.77	0.49	63	0.92	0.78	86	1.16	1.12	96
			BestSub	-0.07	-0.07	100	-0.11	-0.02	16	0.58	0.17	30	0.77	0.49	63	0.92	0.78	86	1.16	1.12	96
		Corr	**LASSO**		0.00		0.47	0.47		0.57	0.57		0.66	0.66		0.81	0.81		0.93	0.93	
			Ridge		0.00		0.48	0.48		0.55	0.55		0.61	0.61		0.72	0.72		0.79	0.79	
			Forward	-0.11	-0.11	100	1.58	0.50	32	1.69	0.60	36	1.70	0.72	42	1.84	0.92	50	1.94	1.10	56
			BestSub	-0.11	-0.11	100	1.58	0.50	32	1.69	0.60	36	1.70	0.72	42	1.84	0.92	50	1.94	1.10	56
MAR	**Low**	**Ind**	**LASSO**		0.00		0.01	0.01		0.29	0.29		0.70	0.70		1.10	1.10		1.50	1.50	
			Ridge		0.00		0.01	0.01		0.30	0.30		0.70	0.70		1.07	1.06		1.43	1.43	
			Forward	-0.01	-0.01	100	0.05	0.01	16	0.61	0.25	41	0.85	0.75	88	1.19	1.18	99	1.59	1.58	100
			BestSub	-0.01	-0.01	100	0.05	0.01	16	0.61	0.25	41	0.85	0.75	88	1.19	1.18	99	1.59	1.58	100
		Corr	**LASSO**		0.00		0.20	0.20		0.29	0.29		0.71	0.71		1.05	1.04		1.31	1.31	
			Ridge		0.00		0.28	0.28		0.37	0.37		0.74	0.74		0.95	0.95		1.09	1.09	
			Forward	-0.65	-0.65	100	0.79	0.17	22	0.97	0.24	25	1.88	0.71	38	2.12	1.15	54	2.33	1.55	67
			BestSub	-0.65	-0.65	100	0.79	0.17	22	0.97	0.24	25	1.88	0.71	38	2.12	1.15	54	2.33	1.55	67
	High	**Ind**	**LASSO**		0.00		0.00	0.00		0.11	0.11		0.27	0.27		0.43	0.43		0.72	0.72	
			Ridge		0.00		-0.00	-0.00		0.11	0.11		0.26	0.26		0.40	0.40		0.62	0.62	
			Forward	-0.09	-0.09	100	0.05	0.01	17	0.59	0.12	20	0.84	0.31	37	0.93	0.54	58	1.15	0.95	83
			BestSub	-0.09	-0.09	100	0.05	0.01	17	0.59	0.12	20	0.84	0.31	37	0.93	0.54	58	1.15	0.95	83
		Corr	**LASSO**		0.00		0.30	0.30		0.38	0.38		0.09	0.09		0.23	0.23		0.48	0.48	
			Ridge		0.00		0.28	0.28		0.35	0.35		0.14	0.14		0.25	0.25		0.39	0.39	
			Forward	-2.27	-2.27	100	1.36	0.39	29	1.69	0.49	29	0.67	0.13	20	1.40	0.31	22	1.67	0.68	40
			BestSub	-2.27	-2.27	100	1.36	0.39	29	1.69	0.49	29	0.67	0.13	20	1.40	0.31	22	1.67	0.68	40

Table 6: **Scenario 2 (No Resampling) | Imputation Method - Deterministic K-Nearest Neighbors Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.46	0.46		0.94	0.94		1.43	1.42		1.92	1.92	
			Ridge		0.00		-0.00	-0.00		0.44	0.44		0.90	0.90		1.35	1.35		1.81	1.81	
			Forward	0.00	0.00	100	-0.01	-0.00	23	0.47	0.47	100	0.95	0.95	100	1.44	1.44	100	1.94	1.93	100
			BestSub	0.00	0.00	100	-0.01	-0.00	23	0.47	0.47	100	0.95	0.95	100	1.44	1.44	100	1.94	1.93	100
		Corr	**LASSO**		0.00		0.08	0.08		0.51	0.51		0.97	0.97		1.45	1.45		1.94	1.94	
			Ridge		0.00		0.25	0.25		0.60	0.60		0.96	0.96		1.34	1.34		1.71	1.71	
			Forward	-0.00	-0.00	100	0.18	0.05	28	0.53	0.52	100	0.98	0.98	100	1.47	1.47	100	1.96	1.95	100
			BestSub	-0.00	-0.00	100	0.18	0.05	28	0.53	0.52	100	0.98	0.98	100	1.47	1.47	100	1.96	1.95	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.41	0.41		0.86	0.85		1.31	1.31		1.81	1.80	
			Ridge		0.00		-0.01	-0.01		0.41	0.41		0.82	0.82		1.25	1.25		1.71	1.71	
			Forward	0.00	0.00	100	-0.01	-0.01	36	0.44	0.43	97	0.87	0.87	100	1.33	1.33	100	1.82	1.82	100
			BestSub	0.00	0.00	100	-0.01	-0.01	36	0.44	0.43	97	0.87	0.87	100	1.33	1.33	100	1.82	1.82	100
		Corr	**LASSO**		0.00		0.21	0.21		0.56	0.56		0.94	0.94		1.38	1.38		1.82	1.82	
			Ridge		0.00		0.36	0.36		0.65	0.65		0.94	0.94		1.28	1.28		1.61	1.61	
			Forward	-0.00	-0.00	100	0.32	0.18	56	0.59	0.58	98	0.96	0.95	100	1.40	1.40	100	1.84	1.84	100
			BestSub	-0.00	-0.00	100	0.32	0.18	56	0.59	0.58	98	0.96	0.95	100	1.40	1.40	100	1.84	1.84	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.46	0.46		0.93	0.93		1.42	1.42		1.91	1.91	
			Ridge		0.00		-0.00	-0.00		0.44	0.44		0.89	0.89		1.35	1.34		1.80	1.80	
			Forward	0.09	0.09	100	-0.00	-0.00	22	0.47	0.47	100	0.95	0.95	100	1.43	1.43	100	1.92	1.92	100
			BestSub	0.09	0.09	100	-0.00	-0.00	22	0.47	0.47	100	0.95	0.95	100	1.43	1.43	100	1.92	1.92	100
		Corr	**LASSO**		0.00		0.09	0.09		0.51	0.51		0.97	0.97		1.45	1.45		1.94	1.94	
			Ridge		0.00		0.25	0.25		0.60	0.60		0.96	0.96		1.34	1.34		1.72	1.71	
			Forward	0.05	0.04	100	0.17	0.05	30	0.53	0.53	100	0.98	0.98	100	1.47	1.46	100	1.96	1.95	100
			BestSub	0.05	0.04	100	0.17	0.05	30	0.53	0.53	100	0.98	0.98	100	1.47	1.46	100	1.96	1.95	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.40	0.40		0.84	0.84		1.29	1.29		1.76	1.76	
			Ridge		0.00		-0.00	-0.00		0.39	0.39		0.81	0.81		1.24	1.23		1.67	1.67	
			Forward	0.15	0.15	100	-0.01	-0.00	33	0.43	0.41	96	0.86	0.86	100	1.31	1.31	100	1.78	1.78	100
			BestSub	0.15	0.15	100	-0.01	-0.00	33	0.43	0.41	96	0.86	0.86	100	1.31	1.31	100	1.78	1.78	100
		Corr	**LASSO**		0.00		0.21	0.21		0.55	0.55		0.95	0.95		1.38	1.38		1.82	1.82	
			Ridge		0.00		0.35	0.35		0.63	0.63		0.95	0.95		1.28	1.28		1.62	1.62	
			Forward	0.10	0.10	100	0.34	0.18	54	0.59	0.56	96	0.97	0.97	100	1.39	1.39	100	1.83	1.83	100
			BestSub	0.10	0.10	100	0.34	0.18	54	0.59	0.56	96	0.97	0.97	100	1.39	1.39	100	1.83	1.83	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.46	0.46		0.92	0.92		1.38	1.38		1.86	1.86	
			Ridge		0.00		-0.00	-0.00		0.45	0.44		0.88	0.87		1.31	1.31		1.76	1.75	
			Forward	0.02	0.02	100	-0.01	-0.00	27	0.47	0.47	100	0.94	0.94	100	1.40	1.40	100	1.88	1.88	100
			BestSub	0.02	0.02	100	-0.01	-0.00	27	0.47	0.47	100	0.94	0.94	100	1.40	1.40	100	1.88	1.88	100
		Corr	**LASSO**		0.00		0.11	0.11		0.52	0.52		0.94	0.94		1.41	1.41		1.86	1.86	
			Ridge		0.00		0.29	0.29		0.62	0.62		0.93	0.93		1.30	1.30		1.65	1.65	
			Forward	-0.06	-0.06	100	0.22	0.08	37	0.54	0.54	99	0.96	0.96	100	1.43	1.43	100	1.88	1.88	100
			BestSub	-0.06	-0.06	100	0.22	0.08	37	0.54	0.54	99	0.96	0.96	100	1.43	1.43	100	1.88	1.88	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.38	0.38		0.79	0.79		1.19	1.19		1.68	1.68	
			Ridge		0.00		-0.01	-0.01		0.37	0.37		0.75	0.75		1.13	1.13		1.58	1.58	
			Forward	-0.01	-0.01	100	-0.00	-0.00	40	0.43	0.39	92	0.80	0.80	100	1.21	1.21	100	1.69	1.69	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	40	0.43	0.39	92	0.80	0.80	100	1.21	1.21	100	1.69	1.69	100
		Corr	**LASSO**		0.00		0.34	0.34		0.61	0.61		0.68	0.68		1.07	1.07		1.51	1.51	
			Ridge		0.00		0.45	0.44		0.67	0.67		0.70	0.70		1.01	1.01		1.35	1.35	
			Forward	-0.50	-0.50	100	0.48	0.31	66	0.68	0.62	91	0.70	0.69	98	1.08	1.08	100	1.52	1.52	100
			BestSub	-0.50	-0.50	100	0.48	0.31	66	0.68	0.62	91	0.70	0.69	98	1.08	1.08	100	1.52	1.52	100

Table 7: **Scenario 2 (No Resampling) | Imputation Method - Stochastic K-Nearest Neighbors Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.43	0.43		0.89	0.89		1.37	1.37		1.87	1.86	
			Ridge		0.00		-0.00	-0.00		0.42	0.42		0.85	0.85		1.30	1.30		1.77	1.76	
			Forward	0.00	0.00	100	-0.01	-0.00	21	0.45	0.45	100	0.91	0.91	100	1.39	1.38	100	1.88	1.88	100
			BestSub	0.00	0.00	100	-0.01	-0.00	21	0.45	0.45	100	0.91	0.91	100	1.39	1.38	100	1.88	1.88	100
		Corr	**LASSO**		0.00		0.12	0.12		0.51	0.51		0.96	0.96		1.43	1.42		1.91	1.91	
			Ridge		0.00		0.27	0.27		0.60	0.60		0.95	0.95		1.32	1.32		1.70	1.70	
			Forward	-0.00	-0.00	100	0.25	0.08	32	0.54	0.53	99	0.98	0.97	100	1.44	1.44	100	1.93	1.93	100
			BestSub	-0.00	-0.00	100	0.25	0.08	32	0.54	0.53	99	0.98	0.97	100	1.44	1.44	100	1.93	1.93	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.35	0.35		0.75	0.75		1.17	1.17		1.64	1.63	
			Ridge		0.00		-0.00	-0.00		0.35	0.35		0.73	0.73		1.13	1.13		1.56	1.56	
			Forward	0.00	0.00	100	-0.02	-0.00	27	0.40	0.36	91	0.77	0.77	100	1.19	1.19	100	1.66	1.65	100
			BestSub	0.00	0.00	100	-0.02	-0.00	27	0.40	0.36	91	0.77	0.77	100	1.19	1.19	100	1.66	1.65	100
		Corr	**LASSO**		0.00		0.29	0.29		0.59	0.59		0.92	0.92		1.33	1.32		1.72	1.72	
			Ridge		0.00		0.40	0.40		0.65	0.65		0.92	0.92		1.24	1.24		1.55	1.55	
			Forward	-0.00	-0.00	100	0.39	0.26	66	0.62	0.60	97	0.94	0.94	100	1.34	1.34	100	1.74	1.74	100
			BestSub	-0.00	-0.00	100	0.39	0.26	66	0.62	0.60	97	0.94	0.94	100	1.34	1.34	100	1.74	1.74	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.43	0.43		0.89	0.89		1.36	1.36		1.85	1.85	
			Ridge		0.00		-0.00	-0.00		0.42	0.42		0.85	0.85		1.30	1.30		1.75	1.75	
			Forward	0.08	0.08	100	-0.01	-0.00	20	0.45	0.44	100	0.90	0.90	100	1.38	1.38	100	1.87	1.87	100
			BestSub	0.08	0.08	100	-0.01	-0.00	20	0.45	0.44	100	0.90	0.90	100	1.38	1.38	100	1.87	1.87	100
		Corr	**LASSO**		0.00		0.12	0.12		0.52	0.52		0.95	0.95		1.42	1.42		1.91	1.91	
			Ridge		0.00		0.27	0.27		0.61	0.61		0.95	0.95		1.32	1.32		1.70	1.70	
			Forward	0.04	0.04	100	0.25	0.09	34	0.54	0.54	99	0.97	0.97	100	1.44	1.43	100	1.92	1.92	100
			BestSub	0.04	0.04	100	0.25	0.09	34	0.54	0.54	99	0.97	0.97	100	1.44	1.43	100	1.92	1.92	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.34	0.34		0.73	0.73		1.14	1.14		1.59	1.59	
			Ridge		0.00		-0.00	-0.00		0.34	0.34		0.71	0.71		1.10	1.10		1.52	1.52	
			Forward	0.07	0.07	100	-0.00	-0.00	28	0.39	0.35	88	0.75	0.75	100	1.16	1.16	100	1.61	1.61	100
			BestSub	0.07	0.07	100	-0.00	-0.00	28	0.39	0.35	88	0.75	0.75	100	1.16	1.16	100	1.61	1.61	100
		Corr	**LASSO**		0.00		0.30	0.30		0.58	0.58		0.92	0.92		1.31	1.30		1.72	1.72	
			Ridge		0.00		0.40	0.40		0.64	0.64		0.92	0.92		1.23	1.23		1.56	1.56	
			Forward	0.10	0.10	100	0.43	0.27	63	0.62	0.59	95	0.94	0.94	100	1.32	1.32	100	1.74	1.74	100
			BestSub	0.10	0.10	100	0.43	0.27	63	0.62	0.59	95	0.94	0.94	100	1.32	1.32	100	1.74	1.74	100
MAR	**Low**	**Ind**	**LASSO**		0.00		0.00	0.00		0.40	0.40		0.92	0.92		1.39	1.39		1.87	1.87	
			Ridge		0.00		0.00	0.00		0.39	0.39		0.88	0.88		1.31	1.31		1.76	1.76	
			Forward	0.02	0.02	100	0.00	0.00	25	0.42	0.42	100	0.94	0.94	100	1.40	1.40	100	1.88	1.88	100
			BestSub	0.02	0.02	100	0.00	0.00	25	0.42	0.42	100	0.94	0.94	100	1.40	1.40	100	1.88	1.88	100
		Corr	**LASSO**		0.00		0.12	0.12		0.46	0.46		0.96	0.96		1.43	1.43		1.89	1.88	
			Ridge		0.00		0.26	0.26		0.56	0.56		0.95	0.95		1.32	1.32		1.68	1.68	
			Forward	-0.06	-0.06	100	0.23	0.08	37	0.48	0.47	97	0.98	0.98	100	1.45	1.45	100	1.90	1.90	100
			BestSub	-0.06	-0.06	100	0.23	0.08	37	0.48	0.47	97	0.98	0.98	100	1.45	1.45	100	1.90	1.90	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.25	0.25		0.79	0.79		1.20	1.20		1.69	1.69	
			Ridge		0.00		-0.01	-0.01		0.25	0.25		0.76	0.76		1.15	1.15		1.60	1.60	
			Forward	-0.06	-0.05	100	-0.02	-0.01	28	0.32	0.25	78	0.81	0.81	100	1.22	1.22	100	1.71	1.71	100
			BestSub	-0.06	-0.05	100	-0.02	-0.01	28	0.32	0.25	78	0.81	0.81	100	1.22	1.22	100	1.71	1.71	100
		Corr	**LASSO**		0.00		0.25	0.25		0.43	0.43		0.73	0.73		1.12	1.12		1.59	1.59	
			Ridge		0.00		0.33	0.33		0.49	0.49		0.76	0.75		1.07	1.07		1.43	1.43	
			Forward	-0.55	-0.55	100	0.40	0.22	55	0.53	0.43	80	0.76	0.74	98	1.14	1.14	100	1.61	1.60	100
			BestSub	-0.55	-0.55	100	0.40	0.22	55	0.53	0.43	80	0.76	0.74	98	1.14	1.14	100	1.61	1.60	100

Table 8: **Scenario 2 (No Resampling) | Imputation Method - Deterministic Regression Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.52	0.52		1.03	1.03		1.53	1.53		2.02	2.02	
			Ridge		0.00		-0.00	-0.00		0.50	0.50		0.98	0.98		1.44	1.44		1.90	1.90	
			Forward	0.00	0.00	100	-0.01	-0.00	38	0.53	0.53	100	1.05	1.04	100	1.54	1.54	100	2.03	2.03	100
			BestSub	0.00	0.00	100	-0.01	-0.00	38	0.53	0.53	100	1.05	1.04	100	1.54	1.54	100	2.03	2.03	100
		Corr	**LASSO**		0.00		-0.01	-0.01		0.47	0.47		1.00	1.00		1.52	1.51		2.02	2.02	
			Ridge		0.00		0.17	0.17		0.59	0.59		0.99	0.99		1.39	1.39		1.78	1.78	
			Forward	0.00	0.00	100	-0.11	-0.04	39	0.49	0.49	99	1.02	1.01	100	1.53	1.53	100	2.04	2.03	100
			BestSub	0.00	0.00	100	-0.11	-0.04	39	0.49	0.49	99	1.02	1.01	100	1.53	1.53	100	2.04	2.03	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.55	0.55		1.07	1.07		1.57	1.57		2.05	2.05	
			Ridge		0.00		-0.01	-0.01		0.52	0.52		1.01	1.01		1.47	1.47		1.92	1.92	
			Forward	-0.00	-0.00	100	-0.02	-0.01	68	0.56	0.56	100	1.08	1.08	100	1.58	1.58	100	2.06	2.06	100
			BestSub	-0.00	-0.00	100	-0.02	-0.01	68	0.56	0.56	100	1.08	1.08	100	1.58	1.58	100	2.06	2.06	100
		Corr	**LASSO**		0.00		-0.05	-0.05		0.47	0.47		1.01	1.01		1.55	1.55		2.05	2.05	
			Ridge		0.00		0.17	0.17		0.60	0.60		1.00	1.00		1.40	1.40		1.78	1.77	
			Forward	-0.00	-0.00	100	-0.15	-0.09	63	0.52	0.48	92	1.03	1.03	100	1.57	1.57	100	2.07	2.07	100
			BestSub	-0.00	-0.00	100	-0.15	-0.09	63	0.52	0.48	92	1.03	1.03	100	1.57	1.57	100	2.07	2.07	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.52	0.52		1.03	1.03		1.52	1.52		2.02	2.01	
			Ridge		0.00		-0.01	-0.01		0.50	0.50		0.98	0.97		1.44	1.43		1.89	1.89	
			Forward	0.08	0.08	100	-0.01	-0.00	36	0.54	0.54	100	1.04	1.04	100	1.54	1.53	100	2.03	2.03	100
			BestSub	0.08	0.08	100	-0.01	-0.00	36	0.54	0.54	100	1.04	1.04	100	1.54	1.53	100	2.03	2.03	100
		Corr	**LASSO**		0.00		-0.01	-0.01		0.48	0.48		1.00	1.00		1.52	1.51		2.02	2.02	
			Ridge		0.00		0.17	0.17		0.59	0.59		0.99	0.99		1.39	1.39		1.78	1.78	
			Forward	0.05	0.05	100	-0.11	-0.04	38	0.50	0.49	99	1.01	1.01	100	1.53	1.53	100	2.03	2.03	100
			BestSub	0.05	0.05	100	-0.11	-0.04	38	0.50	0.49	99	1.01	1.01	100	1.53	1.53	100	2.03	2.03	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.54	0.54		1.06	1.06		1.56	1.56		2.04	2.04	
			Ridge		0.00		-0.01	-0.01		0.51	0.51		1.00	1.00		1.46	1.46		1.91	1.91	
			Forward	0.23	0.23	100	-0.01	-0.01	61	0.55	0.55	99	1.07	1.07	100	1.57	1.57	100	2.05	2.05	100
			BestSub	0.23	0.23	100	-0.01	-0.01	61	0.55	0.55	99	1.07	1.07	100	1.57	1.57	100	2.05	2.05	100
		Corr	**LASSO**		0.00		-0.02	-0.02		0.48	0.47		1.01	1.01		1.54	1.54		2.04	2.04	
			Ridge		0.00		0.19	0.18		0.60	0.60		1.00	1.00		1.40	1.40		1.78	1.78	
			Forward	0.14	0.14	100	-0.11	-0.06	57	0.52	0.49	94	1.03	1.03	100	1.56	1.56	100	2.06	2.06	100
			BestSub	0.14	0.14	100	-0.11	-0.06	57	0.52	0.49	94	1.03	1.03	100	1.56	1.56	100	2.06	2.06	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.56	0.56		0.99	0.99		1.49	1.48		1.99	1.99	
			Ridge		0.00		-0.00	-0.00		0.53	0.53		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	0.03	0.03	100	-0.01	-0.00	37	0.57	0.57	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.03	0.03	100	-0.01	-0.00	37	0.57	0.57	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.02	0.02		0.56	0.56		0.96	0.96		1.47	1.47		1.97	1.97	
			Ridge		0.00		0.22	0.21		0.66	0.66		0.95	0.95		1.35	1.35		1.74	1.73	
			Forward	0.01	0.01	100	-0.04	-0.01	37	0.58	0.58	99	0.98	0.98	100	1.49	1.49	100	1.98	1.98	100
			BestSub	0.01	0.01	100	-0.04	-0.01	37	0.58	0.58	99	0.98	0.98	100	1.49	1.49	100	1.98	1.98	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.58	0.58		0.99	0.99		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.01	-0.01		0.55	0.55		0.93	0.93		1.40	1.40		1.86	1.86	
			Forward	0.07	0.07	100	-0.02	-0.01	58	0.60	0.59	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.07	0.07	100	-0.02	-0.01	58	0.60	0.59	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.04	0.04		0.58	0.58		0.95	0.95		1.47	1.47		1.97	1.96	
			Ridge		0.00		0.31	0.31		0.73	0.73		0.91	0.91		1.32	1.31		1.70	1.69	
			Forward	0.03	0.03	100	-0.03	-0.01	52	0.61	0.59	97	0.97	0.97	100	1.49	1.49	100	1.99	1.98	100
			BestSub	0.03	0.03	100	-0.03	-0.01	52	0.61	0.59	97	0.97	0.97	100	1.49	1.49	100	1.99	1.98	100

Table 9: **Scenario 2 (No Resampling) | Imputation Method - Stochastic Regression Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.48	0.48		0.97	0.97		1.46	1.46		1.95	1.95	
			Ridge		0.00		-0.00	-0.00		0.46	0.46		0.92	0.92		1.38	1.38		1.84	1.84	
			Forward	0.00	0.00	100	-0.01	-0.00	26	0.49	0.49	100	0.98	0.98	100	1.48	1.47	100	1.97	1.97	100
			BestSub	0.00	0.00	100	-0.01	-0.00	26	0.49	0.49	100	0.98	0.98	100	1.48	1.47	100	1.97	1.97	100
		Corr	**LASSO**		0.00		0.05	0.05		0.49	0.49		0.98	0.98		1.48	1.48		1.97	1.97	
			Ridge		0.00		0.21	0.21		0.59	0.59		0.97	0.97		1.36	1.36		1.75	1.75	
			Forward	-0.00	-0.00	100	0.06	0.02	30	0.51	0.50	98	0.99	0.99	100	1.49	1.49	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	0.06	0.02	30	0.51	0.50	98	0.99	0.99	100	1.49	1.49	100	1.99	1.99	100
	High	**Ind**	**LASSO**		0.00		-0.02	-0.02		0.42	0.42		0.88	0.87		1.34	1.34		1.81	1.81	
			Ridge		0.00		-0.01	-0.01		0.41	0.41		0.84	0.84		1.27	1.27		1.71	1.71	
			Forward	-0.00	-0.00	100	-0.04	-0.01	36	0.45	0.43	97	0.89	0.89	100	1.35	1.35	100	1.83	1.82	100
			BestSub	-0.00	-0.00	100	-0.04	-0.01	36	0.45	0.43	97	0.89	0.89	100	1.35	1.35	100	1.83	1.82	100
		Corr	**LASSO**		0.00		0.13	0.13		0.53	0.52		0.95	0.95		1.43	1.43		1.89	1.89	
			Ridge		0.00		0.28	0.28		0.61	0.61		0.95	0.95		1.32	1.32		1.68	1.68	
			Forward	-0.01	-0.01	100	0.23	0.10	44	0.56	0.54	95	0.97	0.97	100	1.44	1.44	100	1.91	1.90	100
			BestSub	-0.01	-0.01	100	0.23	0.10	44	0.56	0.54	95	0.97	0.97	100	1.44	1.44	100	1.91	1.90	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.48	0.48		0.97	0.97		1.46	1.46		1.95	1.95	
			Ridge		0.00		-0.01	-0.01		0.46	0.46		0.92	0.92		1.38	1.38		1.84	1.83	
			Forward	0.07	0.07	100	-0.00	-0.00	28	0.49	0.49	100	0.98	0.98	100	1.47	1.47	100	1.96	1.96	100
			BestSub	0.07	0.07	100	-0.00	-0.00	28	0.49	0.49	100	0.98	0.98	100	1.47	1.47	100	1.96	1.96	100
		Corr	**LASSO**		0.00		0.05	0.05		0.49	0.49		0.98	0.98		1.47	1.47		1.97	1.97	
			Ridge		0.00		0.21	0.21		0.59	0.59		0.97	0.97		1.36	1.36		1.75	1.75	
			Forward	0.05	0.05	100	0.07	0.02	30	0.51	0.50	99	0.99	0.99	100	1.49	1.49	100	1.99	1.98	100
			BestSub	0.05	0.05	100	0.07	0.02	30	0.51	0.50	99	0.99	0.99	100	1.49	1.49	100	1.99	1.98	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.41	0.41		0.86	0.86		1.32	1.32		1.78	1.77	
			Ridge		0.00		-0.00	-0.00		0.40	0.40		0.83	0.83		1.26	1.25		1.68	1.68	
			Forward	0.13	0.13	100	-0.00	-0.00	33	0.44	0.42	96	0.88	0.88	100	1.34	1.33	100	1.79	1.79	100
			BestSub	0.13	0.13	100	-0.00	-0.00	33	0.44	0.42	96	0.88	0.88	100	1.34	1.33	100	1.79	1.79	100
		Corr	**LASSO**		0.00		0.16	0.16		0.54	0.54		0.95	0.95		1.41	1.41		1.86	1.86	
			Ridge		0.00		0.30	0.30		0.62	0.62		0.95	0.95		1.31	1.31		1.67	1.67	
			Forward	0.14	0.14	100	0.30	0.13	44	0.58	0.55	95	0.97	0.97	100	1.43	1.43	100	1.88	1.88	100
			BestSub	0.14	0.14	100	0.30	0.13	44	0.58	0.55	95	0.97	0.97	100	1.43	1.43	100	1.88	1.88	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.46	0.46		0.99	0.99		1.48	1.48		1.99	1.99	
			Ridge		0.00		-0.00	-0.00		0.44	0.44		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	0.02	0.02	100	-0.02	-0.01	30	0.48	0.47	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.02	0.02	100	-0.02	-0.01	30	0.48	0.47	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.04	0.04		0.45	0.45		0.99	0.99		1.50	1.50		1.99	1.99	
			Ridge		0.00		0.20	0.20		0.56	0.56		0.98	0.98		1.38	1.38		1.77	1.77	
			Forward	0.01	0.01	100	0.04	0.01	30	0.48	0.47	98	1.00	1.00	100	1.51	1.51	100	2.01	2.01	100
			BestSub	0.01	0.01	100	0.04	0.01	30	0.48	0.47	98	1.00	1.00	100	1.51	1.51	100	2.01	2.01	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.31	0.31		0.98	0.98		1.49	1.49		1.99	1.98	
			Ridge		0.00		-0.01	-0.01		0.30	0.30		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	-0.03	-0.03	100	-0.01	-0.00	33	0.33	0.32	97	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.03	-0.03	100	-0.01	-0.00	33	0.33	0.32	97	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.05	0.05		0.29	0.29		1.01	1.01		1.53	1.53		2.03	2.03	
			Ridge		0.00		0.22	0.22		0.43	0.43		1.00	1.00		1.41	1.40		1.80	1.79	
			Forward	-0.02	-0.02	100	0.08	0.02	30	0.35	0.30	85	1.03	1.03	100	1.55	1.54	100	2.05	2.05	100
			BestSub	-0.02	-0.02	100	0.08	0.02	30	0.35	0.30	85	1.03	1.03	100	1.55	1.54	100	2.05	2.05	100

Table 10: **Scenario 3 (No Imputation and No Resampling)**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00	0	-0.00	-0.00	100	0.48	0.48	100	0.99	0.98	100	1.48	1.48	100	1.98	1.98	100
			Ridge		0.00	0	-0.00	-0.00	100	0.47	0.47	100	0.94	0.94	100	1.40	1.40	100	1.87	1.87	100
			Forward	0.00	0.00	100	-0.01	-0.00	16	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.01	-0.00	16	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00	0	0.04	0.04	100	0.48	0.48	100	0.97	0.97	100	1.48	1.48	100	1.99	1.99	100
			Ridge		0.00	0	0.20	0.20	100	0.58	0.58	100	0.97	0.97	100	1.37	1.37	100	1.76	1.76	100
			Forward	0.00	0.00	100	0.04	0.01	17	0.51	0.49	95	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
			BestSub	0.00	0.00	100	0.04	0.01	17	0.51	0.49	95	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**LASSO**		0.00	0	-0.01	-0.01	100	0.46	0.46	100	0.96	0.96	100	1.46	1.46	100	1.95	1.95	100
			Ridge		0.00	0	-0.01	-0.01	100	0.46	0.46	100	0.93	0.93	100	1.39	1.39	100	1.84	1.84	100
			Forward	-0.00	-0.00	100	-0.05	-0.01	21	0.53	0.49	91	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.05	-0.01	21	0.53	0.49	91	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**LASSO**		0.00	0	0.07	0.07	100	0.48	0.48	100	0.95	0.95	100	1.48	1.48	100	1.97	1.97	100
			Ridge		0.00	0	0.23	0.23	100	0.61	0.60	100	0.96	0.96	100	1.36	1.36	100	1.73	1.73	100
			Forward	-0.00	-0.00	100	0.08	0.02	19	0.65	0.45	68	1.01	0.99	98	1.53	1.53	100	2.02	2.02	100
			BestSub	-0.00	-0.00	100	0.08	0.02	19	0.65	0.45	68	1.01	0.99	98	1.53	1.53	100	2.02	2.02	100
MNAR	**Low**	**Ind**	**LASSO**		0.00	0	-0.00	-0.00	100	0.48	0.48	100	0.98	0.98	100	1.48	1.48	100	1.98	1.97	100
			Ridge		0.00	0	-0.00	-0.00	100	0.47	0.47	100	0.94	0.94	100	1.40	1.40	100	1.86	1.86	100
			Forward	-0.02	-0.02	100	-0.00	-0.00	16	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.00	-0.00	16	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**LASSO**		0.00	0	0.04	0.04	100	0.48	0.48	100	0.97	0.97	100	1.48	1.48	100	1.98	1.98	100
			Ridge		0.00	0	0.19	0.19	100	0.58	0.58	100	0.97	0.97	100	1.37	1.37	100	1.77	1.76	100
			Forward	-0.01	-0.01	100	0.00	0.00	18	0.51	0.50	97	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	0.00	0.00	18	0.51	0.50	97	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**LASSO**		0.00	0	-0.01	-0.01	100	0.46	0.46	100	0.95	0.95	100	1.45	1.45	100	1.94	1.94	100
			Ridge		0.00	0	-0.01	-0.01	100	0.47	0.47	100	0.92	0.92	100	1.39	1.38	100	1.84	1.83	100
			Forward	-0.07	-0.07	100	-0.02	-0.00	17	0.53	0.49	92	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.02	-0.00	17	0.53	0.49	92	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**LASSO**		0.00	0	0.06	0.06	100	0.48	0.48	100	0.96	0.96	100	1.48	1.47	100	1.96	1.96	100
			Ridge		0.00	0	0.17	0.17	100	0.57	0.57	100	0.96	0.96	100	1.38	1.38	100	1.78	1.78	100
			Forward	-0.04	-0.04	100	0.08	0.02	19	0.59	0.46	78	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
			BestSub	-0.04	-0.04	100	0.08	0.02	19	0.59	0.46	78	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
MAR	**Low**	**Ind**	**LASSO**		0.00	0	-0.00	-0.00	100	0.49	0.49	100	0.98	0.98	100	1.48	1.48	100	1.98	1.98	100
			Ridge		0.00	0	-0.00	-0.00	100	0.47	0.47	100	0.93	0.93	100	1.40	1.40	100	1.87	1.87	100
			Forward	0.00	0.00	100	-0.02	-0.00	18	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.02	-0.00	18	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00	0	0.03	0.03	100	0.49	0.49	100	0.98	0.98	100	1.49	1.49	100	1.98	1.98	100
			Ridge		0.00	0	0.17	0.17	100	0.58	0.58	100	0.97	0.97	100	1.39	1.39	100	1.78	1.78	100
			Forward	-0.00	-0.00	100	0.00	0.00	19	0.51	0.50	97	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
			BestSub	-0.00	-0.00	100	0.00	0.00	19	0.51	0.50	97	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**LASSO**		0.00	0	-0.01	-0.01	100	0.47	0.47	100	0.97	0.96	100	1.46	1.46	100	1.98	1.97	100
			Ridge		0.00	0	-0.01	-0.01	100	0.46	0.46	100	0.93	0.92	100	1.39	1.38	100	1.86	1.86	100
			Forward	-0.02	-0.02	100	-0.01	-0.00	18	0.52	0.49	94	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.02	-0.02	100	-0.01	-0.00	18	0.52	0.49	94	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**LASSO**		0.00	0	0.04	0.04	100	0.47	0.47	100	0.96	0.96	100	1.47	1.47	100	1.98	1.98	100
			Ridge		0.00	0	0.12	0.12	100	0.53	0.53	100	0.96	0.96	100	1.41	1.41	100	1.82	1.82	100
			Forward	-0.02	-0.02	100	0.04	0.01	19	0.56	0.46	82	1.00	0.99	100	1.51	1.51	100	2.01	2.01	100
			BestSub	-0.02	-0.02	100	0.04	0.01	19	0.56	0.46	82	1.00	0.99	100	1.51	1.51	100	2.01	2.01	100

Table 11: **Scenario 4 (Imputation and Resampling) | Combination Method - Stability Selection | Imputation Method - Deterministic Mean Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.02	-0.00	16	0.50	0.50	99	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.06	0.03	43	0.50	0.47	93	1.00	1.00	100	1.51	1.50	100	2.01	2.01	100
			BestSub	0.00	0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.03	-0.01	20	0.52	0.45	87	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.07	0.04	64	0.52	0.45	86	1.00	0.97	97	1.53	1.52	100	2.02	2.02	100
			BestSub	-0.00	-0.00	100	0.01	0.01	100	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	0.00	0.00	16	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	0.06	0.03	43	0.50	0.47	94	1.00	0.99	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.01	-0.00	17	0.51	0.44	86	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.04	-0.04	100	0.06	0.04	62	0.53	0.46	87	1.00	0.97	97	1.51	1.50	99	2.01	2.01	100
			BestSub	-0.04	-0.04	100	0.01	0.01	100	0.50	0.50	100	0.99	0.98	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	0.01	0.00	16	0.51	0.50	98	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	-0.01	-0.01	100	0.09	0.03	36	0.54	0.35	65	1.04	1.01	97	1.55	1.55	100	2.04	2.04	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.05	-0.05	100	-0.01	-0.00	17	0.53	0.39	73	0.99	0.97	98	1.49	1.49	100	1.99	1.99	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.64	-0.64	100	0.37	0.36	97	0.73	0.73	99	1.14	0.17	15	1.66	0.80	48	2.35	2.33	99
			BestSub	-0.00	-0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100

Table 12: **Scenario 4 (Imputation and Resampling) | Combination Method - Stacking Selection | Imputation Method - Deterministic Mean Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.44	0.44		0.90	0.90		1.35	1.35		1.79	1.79	
			Ridge		0.00		-0.01	-0.01		0.43	0.43		0.85	0.85		1.27	1.27		1.70	1.69	
			Forward	0.00	0.00	100	-0.01	-0.01	90	0.45	0.45	100	0.91	0.90	100	1.35	1.35	100	1.80	1.80	100
			BestSub	0.00	0.00	100	-0.01	-0.01	90	0.45	0.45	100	0.91	0.90	100	1.35	1.35	100	1.80	1.80	100
		Corr	**LASSO**		0.00		0.22	0.22		0.59	0.59		0.97	0.97		1.33	1.33		1.71	1.71	
			Ridge		0.00		0.33	0.33		0.64	0.64		0.95	0.95		1.25	1.25		1.57	1.57	
			Forward	-0.00	-0.00	100	0.24	0.22	94	0.60	0.60	99	0.97	0.97	100	1.34	1.34	100	1.72	1.72	100
			BestSub	-0.00	-0.00	100	0.24	0.22	94	0.60	0.60	99	0.97	0.97	100	1.34	1.34	100	1.72	1.72	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.37	0.37		0.74	0.74		1.11	1.11		1.49	1.49	
			Ridge		0.00		-0.01	-0.01		0.36	0.36		0.71	0.71		1.07	1.06		1.42	1.42	
			Forward	0.00	0.00	100	-0.01	-0.01	91	0.38	0.38	100	0.75	0.75	100	1.12	1.12	100	1.50	1.50	100
			BestSub	0.00	0.00	100	-0.01	-0.01	91	0.38	0.38	100	0.75	0.75	100	1.12	1.12	100	1.50	1.50	100
		Corr	**LASSO**		0.00		0.43	0.43		0.66	0.66		0.91	0.91		1.14	1.14		1.41	1.41	
			Ridge		0.00		0.47	0.47		0.67	0.67		0.89	0.89		1.10	1.10		1.33	1.33	
			Forward	-0.01	-0.01	100	0.45	0.43	97	0.67	0.66	100	0.92	0.91	100	1.15	1.15	100	1.42	1.42	100
			BestSub	-0.01	-0.01	100	0.45	0.43	97	0.67	0.66	100	0.92	0.91	100	1.15	1.15	100	1.42	1.42	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.44	0.44		0.89	0.89		1.34	1.34		1.79	1.78	
			Ridge		0.00		-0.00	-0.00		0.42	0.42		0.85	0.84		1.27	1.27		1.69	1.69	
			Forward	0.10	0.10	100	-0.00	-0.00	89	0.45	0.45	100	0.90	0.90	100	1.35	1.34	100	1.79	1.79	100
			BestSub	0.10	0.10	100	-0.00	-0.00	89	0.45	0.45	100	0.90	0.90	100	1.35	1.34	100	1.79	1.79	100
		Corr	**LASSO**		0.00		0.22	0.22		0.58	0.58		0.97	0.97		1.34	1.34		1.72	1.72	
			Ridge		0.00		0.32	0.32		0.63	0.63		0.95	0.95		1.26	1.26		1.58	1.58	
			Forward	0.03	0.03	100	0.23	0.22	92	0.60	0.59	99	0.98	0.97	100	1.35	1.35	100	1.73	1.73	100
			BestSub	0.03	0.03	100	0.23	0.22	92	0.60	0.59	99	0.98	0.97	100	1.35	1.35	100	1.73	1.73	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.36	0.36		0.73	0.73		1.10	1.10		1.48	1.48	
			Ridge		0.00		-0.00	-0.00		0.35	0.35		0.70	0.70		1.05	1.05		1.41	1.41	
			Forward	0.15	0.15	100	-0.00	-0.00	88	0.37	0.36	99	0.74	0.74	100	1.11	1.11	100	1.48	1.48	100
			BestSub	0.15	0.15	100	-0.00	-0.00	88	0.37	0.36	99	0.74	0.74	100	1.11	1.11	100	1.48	1.48	100
		Corr	**LASSO**		0.00		0.42	0.42		0.65	0.65		0.91	0.91		1.18	1.18		1.44	1.44	
			Ridge		0.00		0.46	0.46		0.67	0.67		0.90	0.90		1.14	1.14		1.37	1.36	
			Forward	0.04	0.04	100	0.44	0.42	96	0.66	0.65	99	0.92	0.92	100	1.19	1.19	100	1.45	1.45	100
			BestSub	0.04	0.04	100	0.44	0.42	96	0.66	0.65	99	0.92	0.92	100	1.19	1.19	100	1.45	1.45	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.40	0.40		0.79	0.79		1.18	1.18		1.59	1.59	
			Ridge		0.00		-0.00	-0.00		0.39	0.39		0.75	0.75		1.12	1.11		1.50	1.50	
			Forward	0.00	0.00	100	-0.00	-0.00	86	0.41	0.41	100	0.80	0.80	100	1.18	1.18	100	1.60	1.59	100
			BestSub	0.00	0.00	100	-0.00	-0.00	86	0.41	0.41	100	0.80	0.80	100	1.18	1.18	100	1.60	1.59	100
		Corr	**LASSO**		0.00		0.15	0.15		0.39	0.39		0.74	0.74		1.14	1.14		1.57	1.56	
			Ridge		0.00		0.24	0.24		0.45	0.45		0.76	0.75		1.08	1.08		1.42	1.42	
			Forward	-0.63	-0.63	100	0.16	0.15	92	0.40	0.39	97	0.75	0.74	100	1.15	1.15	100	1.57	1.57	100
			BestSub	-0.63	-0.63	100	0.16	0.15	92	0.40	0.39	97	0.75	0.74	100	1.15	1.15	100	1.57	1.57	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.34	0.34		0.43	0.43		0.65	0.65		0.98	0.98	
			Ridge		0.00		-0.01	-0.01		0.33	0.33		0.41	0.41		0.62	0.62		0.94	0.94	
			Forward	-0.02	-0.02	100	-0.01	-0.01	88	0.35	0.34	98	0.43	0.43	100	0.65	0.65	100	0.99	0.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	88	0.35	0.34	98	0.43	0.43	100	0.65	0.65	100	0.99	0.99	100
		Corr	**LASSO**		0.00		0.88	0.88		1.02	1.02		0.01	0.01		0.26	0.26		0.74	0.74	
			Ridge		0.00		0.86	0.86		0.99	0.99		0.05	0.05		0.27	0.27		0.69	0.69	
			Forward	-1.72	-1.72	100	0.88	0.88	100	1.02	1.02	100	0.01	0.01	89	0.28	0.27	95	0.75	0.75	100
			BestSub	-1.72	-1.72	100	0.88	0.88	100	1.02	1.02	100	0.01	0.01	89	0.28	0.27	95	0.75	0.75	100

Table 13: **Scenario 4 (Imputation and Resampling) | Combination Method - Stability Selection | Imputation Method - Stochastic Mean Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.00	-0.00	15	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.13	0.04	34	0.49	0.49	100	0.99	0.98	100	1.49	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.02	-0.00	23	0.51	0.49	95	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.07	0.05	73	0.50	0.49	99	0.98	0.97	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.00	-0.00	100	0.01	0.01	100	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	-0.01	-0.00	15	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	0.12	0.04	34	0.49	0.49	100	0.98	0.98	100	1.49	1.49	100	1.99	1.99	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.02	-0.00	22	0.51	0.48	94	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.03	-0.03	100	0.08	0.05	68	0.50	0.49	99	0.98	0.98	100	1.49	1.49	100	1.99	1.98	100
			BestSub	-0.04	-0.04	100	0.01	0.01	100	0.50	0.50	100	0.99	0.98	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	0.00	0.00	20	0.51	0.50	100	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	-0.00	-0.00	100	0.12	0.04	32	0.50	0.49	99	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.02	-0.02	100	-0.02	-0.00	19	0.53	0.45	86	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.01	-0.01	100	0.05	0.04	65	0.52	0.46	89	1.00	0.99	99	1.50	1.50	100	2.01	2.01	100
			BestSub	-0.00	-0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100

Table 14: **Scenario 4 (Imputation and Resampling) | Combination Method - Stacking Selection | Imputation Method - Stochastic Mean Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.41	0.41		0.80	0.80		1.20	1.20		1.60	1.60	
			Ridge		0.00		-0.01	-0.01		0.39	0.39		0.77	0.77		1.14	1.14		1.52	1.52	
			Forward	0.04	0.04	100	-0.01	-0.01	85	0.42	0.42	100	0.81	0.81	100	1.21	1.21	100	1.61	1.61	100
			BestSub	0.04	0.04	100	-0.01	-0.01	85	0.42	0.42	100	0.81	0.81	100	1.21	1.21	100	1.61	1.61	100
		Corr	**LASSO**		0.00		0.36	0.36		0.67	0.67		0.93	0.93		1.20	1.20		1.50	1.50	
			Ridge		0.00		0.43	0.43		0.70	0.69		0.92	0.92		1.15	1.15		1.40	1.40	
			Forward	0.05	0.05	100	0.37	0.37	98	0.68	0.68	100	0.94	0.94	100	1.21	1.21	100	1.50	1.50	100
			BestSub	0.05	0.05	100	0.37	0.37	98	0.68	0.68	100	0.94	0.94	100	1.21	1.21	100	1.50	1.50	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.27	0.27		0.54	0.54		0.82	0.82		1.11	1.11	
			Ridge		0.00		-0.01	-0.01		0.27	0.27		0.53	0.52		0.79	0.79		1.07	1.07	
			Forward	0.11	0.11	100	-0.01	-0.01	84	0.28	0.28	99	0.55	0.55	100	0.83	0.83	100	1.11	1.11	100
			BestSub	0.11	0.11	100	-0.01	-0.01	84	0.28	0.28	99	0.55	0.55	100	0.83	0.83	100	1.11	1.11	100
		Corr	**LASSO**		0.00		0.55	0.55		0.67	0.67		0.79	0.79		0.94	0.94		1.11	1.11	
			Ridge		0.00		0.56	0.56		0.67	0.67		0.78	0.78		0.92	0.92		1.08	1.08	
			Forward	0.14	0.14	100	0.56	0.55	100	0.68	0.68	100	0.80	0.80	100	0.95	0.95	100	1.12	1.12	100
			BestSub	0.14	0.14	100	0.56	0.55	100	0.68	0.68	100	0.80	0.80	100	0.95	0.95	100	1.12	1.12	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.41	0.41		0.80	0.80		1.21	1.21		1.61	1.61	
			Ridge		0.00		-0.00	-0.00		0.39	0.39		0.77	0.77		1.15	1.15		1.53	1.53	
			Forward	0.11	0.11	100	-0.00	-0.00	85	0.42	0.42	100	0.81	0.81	100	1.22	1.22	100	1.62	1.62	100
			BestSub	0.11	0.11	100	-0.00	-0.00	85	0.42	0.42	100	0.81	0.81	100	1.22	1.22	100	1.62	1.62	100
		Corr	**LASSO**		0.00		0.36	0.36		0.67	0.67		0.93	0.93		1.23	1.23		1.53	1.53	
			Ridge		0.00		0.43	0.43		0.70	0.70		0.92	0.92		1.18	1.18		1.43	1.43	
			Forward	0.08	0.08	100	0.37	0.36	99	0.68	0.68	100	0.94	0.94	100	1.24	1.24	100	1.53	1.53	100
			BestSub	0.08	0.08	100	0.37	0.36	99	0.68	0.68	100	0.94	0.94	100	1.24	1.24	100	1.53	1.53	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.27	0.27		0.54	0.54		0.82	0.82		1.10	1.10	
			Ridge		0.00		-0.00	-0.00		0.27	0.27		0.53	0.52		0.79	0.79		1.06	1.06	
			Forward	0.08	0.08	100	-0.00	-0.00	83	0.28	0.28	99	0.55	0.55	100	0.83	0.82	100	1.10	1.10	100
			BestSub	0.08	0.08	100	-0.00	-0.00	83	0.28	0.28	99	0.55	0.55	100	0.83	0.82	100	1.10	1.10	100
		Corr	**LASSO**		0.00		0.55	0.55		0.71	0.71		0.83	0.82		0.98	0.98		1.13	1.13	
			Ridge		0.00		0.56	0.56		0.71	0.71		0.81	0.81		0.96	0.96		1.10	1.10	
			Forward	0.11	0.11	100	0.56	0.56	100	0.71	0.71	100	0.83	0.83	100	0.99	0.99	100	1.14	1.14	100
			BestSub	0.11	0.11	100	0.56	0.56	100	0.71	0.71	100	0.83	0.83	100	0.99	0.99	100	1.14	1.14	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.33	0.33		0.79	0.79		1.18	1.18		1.59	1.59	
			Ridge		0.00		-0.00	-0.00		0.32	0.32		0.75	0.75		1.12	1.12		1.51	1.51	
			Forward	0.05	0.05	100	-0.00	-0.00	82	0.33	0.33	100	0.80	0.80	100	1.18	1.18	100	1.60	1.59	100
			BestSub	0.05	0.05	100	-0.00	-0.00	82	0.33	0.33	100	0.80	0.80	100	1.18	1.18	100	1.60	1.59	100
		Corr	**LASSO**		0.00		0.13	0.13		0.27	0.27		0.79	0.79		1.20	1.20		1.60	1.60	
			Ridge		0.00		0.19	0.19		0.32	0.32		0.81	0.81		1.15	1.15		1.47	1.47	
			Forward	-0.57	-0.57	100	0.14	0.13	88	0.29	0.28	95	0.80	0.80	100	1.21	1.21	100	1.61	1.61	100
			BestSub	-0.57	-0.57	100	0.14	0.13	88	0.29	0.28	95	0.80	0.80	100	1.21	1.21	100	1.61	1.61	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.19	0.19		0.43	0.43		0.64	0.64		0.99	0.99	
			Ridge		0.00		-0.00	-0.00		0.18	0.18		0.42	0.42		0.62	0.62		0.95	0.95	
			Forward	0.11	0.11	100	-0.00	-0.00	76	0.19	0.19	98	0.44	0.44	100	0.65	0.65	100	0.99	0.99	100
			BestSub	0.11	0.11	100	-0.00	-0.00	76	0.19	0.19	98	0.44	0.44	100	0.65	0.65	100	0.99	0.99	100
		Corr	**LASSO**		0.00		0.48	0.48		0.54	0.54		0.12	0.12		0.38	0.38		0.87	0.86	
			Ridge		0.00		0.47	0.47		0.53	0.53		0.15	0.15		0.39	0.39		0.82	0.82	
			Forward	-2.01	-2.01	100	0.48	0.48	100	0.54	0.54	100	0.14	0.12	85	0.39	0.38	98	0.87	0.87	100
			BestSub	-2.01	-2.01	100	0.48	0.48	100	0.54	0.54	100	0.14	0.12	85	0.39	0.38	98	0.87	0.87	100

Table 15: **Scenario 4 (Imputation and Resampling) | Combination Method - Stability Selection Imputation Method - Deterministic K-Nearest Neighbors Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.01	-0.00	22	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.11	0.04	32	0.49	0.49	100	0.99	0.99	100	1.50	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**Forward**	-0.01	-0.01	100	-0.02	-0.01	36	0.51	0.49	97	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.09	0.06	64	0.50	0.49	99	0.98	0.98	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.00	-0.00	100	0.01	0.01	100	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	-0.01	-0.00	22	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	0.10	0.03	32	0.49	0.49	100	0.99	0.99	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.01	-0.00	34	0.51	0.49	97	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.03	-0.03	100	0.08	0.05	63	0.50	0.49	98	0.98	0.98	100	1.49	1.49	100	1.99	1.98	100
			BestSub	-0.04	-0.04	100	0.01	0.01	100	0.50	0.50	100	0.99	0.98	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.00	-0.00	27	0.51	0.50	100	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.09	0.04	39	0.50	0.49	100	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.01	-0.01	100	-0.01	-0.01	45	0.51	0.48	94	1.00	0.99	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.01	-0.01	100	0.03	0.02	81	0.51	0.48	95	1.00	0.98	99	1.50	1.50	100	2.01	2.01	100
			BestSub	-0.00	-0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100

Table 16: **Scenario 4 (Imputation and Resampling) | Combination Method - Stacking Selection | Imputation Method - Deterministic K-Nearest Neighbors Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.46	0.46		0.93	0.93		1.41	1.41		1.91	1.91	
			Ridge		0.00		-0.00	-0.00		0.44	0.44		0.88	0.88		1.33	1.33		1.80	1.79	
			Forward	0.00	0.00	100	-0.01	-0.00	91	0.46	0.46	100	0.94	0.94	100	1.42	1.42	100	1.92	1.92	100
			BestSub	0.00	0.00	100	-0.01	-0.00	91	0.46	0.46	100	0.94	0.94	100	1.42	1.42	100	1.92	1.92	100
		Corr	**LASSO**		0.00		0.09	0.09		0.51	0.51		0.97	0.96		1.44	1.44		1.93	1.93	
			Ridge		0.00		0.26	0.26		0.60	0.60		0.96	0.96		1.33	1.33		1.71	1.71	
			Forward	-0.00	-0.00	100	0.10	0.09	90	0.52	0.52	100	0.98	0.97	100	1.45	1.45	100	1.94	1.94	100
			BestSub	-0.00	-0.00	100	0.10	0.09	90	0.52	0.52	100	0.98	0.97	100	1.45	1.45	100	1.94	1.94	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.40	0.40		0.83	0.83		1.28	1.28		1.76	1.76	
			Ridge		0.00		-0.01	-0.01		0.39	0.39		0.79	0.79		1.21	1.21		1.67	1.66	
			Forward	0.00	0.00	100	-0.01	-0.01	93	0.41	0.41	100	0.84	0.84	100	1.29	1.28	100	1.77	1.77	100
			BestSub	0.00	0.00	100	-0.01	-0.01	93	0.41	0.41	100	0.84	0.84	100	1.29	1.28	100	1.77	1.77	100
		Corr	**LASSO**		0.00		0.23	0.23		0.58	0.58		0.94	0.94		1.36	1.36		1.79	1.79	
			Ridge		0.00		0.37	0.37		0.65	0.65		0.94	0.94		1.26	1.26		1.59	1.59	
			Forward	-0.00	-0.00	100	0.24	0.24	97	0.59	0.59	100	0.94	0.94	100	1.36	1.36	100	1.80	1.79	100
			BestSub	-0.00	-0.00	100	0.24	0.24	97	0.59	0.59	100	0.94	0.94	100	1.36	1.36	100	1.80	1.79	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.45	0.45		0.92	0.92		1.41	1.41		1.90	1.89	
			Ridge		0.00		-0.00	-0.00		0.43	0.43		0.88	0.88		1.33	1.33		1.78	1.78	
			Forward	0.08	0.08	100	-0.00	-0.00	90	0.46	0.46	100	0.93	0.93	100	1.42	1.41	100	1.90	1.90	100
			BestSub	0.08	0.08	100	-0.00	-0.00	90	0.46	0.46	100	0.93	0.93	100	1.42	1.41	100	1.90	1.90	100
		Corr	**LASSO**		0.00		0.09	0.09		0.52	0.52		0.97	0.96		1.44	1.44		1.93	1.93	
			Ridge		0.00		0.26	0.26		0.61	0.61		0.96	0.96		1.33	1.33		1.71	1.71	
			Forward	0.04	0.04	100	0.09	0.09	90	0.53	0.53	100	0.98	0.97	100	1.45	1.45	100	1.94	1.94	100
			BestSub	0.04	0.04	100	0.09	0.09	90	0.53	0.53	100	0.98	0.97	100	1.45	1.45	100	1.94	1.94	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.39	0.39		0.81	0.81		1.26	1.26		1.72	1.72	
			Ridge		0.00		-0.00	-0.00		0.38	0.37		0.78	0.77		1.19	1.19		1.62	1.62	
			Forward	0.12	0.12	100	-0.00	-0.00	90	0.40	0.40	100	0.82	0.82	100	1.26	1.26	100	1.73	1.72	100
			BestSub	0.12	0.12	100	-0.00	-0.00	90	0.40	0.40	100	0.82	0.82	100	1.26	1.26	100	1.73	1.72	100
		Corr	**LASSO**		0.00		0.24	0.24		0.57	0.57		0.94	0.94		1.35	1.35		1.78	1.78	
			Ridge		0.00		0.37	0.37		0.64	0.64		0.94	0.94		1.27	1.26		1.60	1.60	
			Forward	0.10	0.10	100	0.26	0.24	95	0.58	0.58	100	0.95	0.95	100	1.36	1.36	100	1.79	1.79	100
			BestSub	0.10	0.10	100	0.26	0.24	95	0.58	0.58	100	0.95	0.95	100	1.36	1.36	100	1.79	1.79	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.44	0.44		0.92	0.92		1.38	1.38		1.86	1.86	
			Ridge		0.00		-0.00	-0.00		0.43	0.43		0.87	0.87		1.30	1.30		1.75	1.75	
			Forward	0.02	0.02	100	-0.00	-0.00	92	0.45	0.45	100	0.93	0.93	100	1.39	1.39	100	1.87	1.87	100
			BestSub	0.02	0.02	100	-0.00	-0.00	92	0.45	0.45	100	0.93	0.93	100	1.39	1.39	100	1.87	1.87	100
		Corr	**LASSO**		0.00		0.11	0.11		0.51	0.51		0.95	0.95		1.41	1.41		1.86	1.86	
			Ridge		0.00		0.28	0.28		0.60	0.60		0.94	0.94		1.30	1.30		1.65	1.65	
			Forward	-0.07	-0.07	100	0.12	0.11	92	0.52	0.51	100	0.96	0.96	100	1.42	1.42	100	1.87	1.87	100
			BestSub	-0.07	-0.07	100	0.12	0.11	92	0.52	0.51	100	0.96	0.96	100	1.42	1.42	100	1.87	1.87	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.35	0.35		0.79	0.79		1.18	1.18		1.67	1.67	
			Ridge		0.00		-0.00	-0.00		0.34	0.34		0.74	0.74		1.12	1.12		1.57	1.57	
			Forward	-0.02	-0.02	100	-0.01	-0.00	93	0.36	0.36	100	0.79	0.79	100	1.19	1.19	100	1.67	1.67	100
			BestSub	-0.02	-0.02	100	-0.01	-0.00	93	0.36	0.36	100	0.79	0.79	100	1.19	1.19	100	1.67	1.67	100
		Corr	**LASSO**		0.00		0.31	0.31		0.55	0.55		0.68	0.67		1.07	1.06		1.50	1.50	
			Ridge		0.00		0.41	0.41		0.61	0.61		0.70	0.70		1.01	1.01		1.35	1.35	
			Forward	-0.56	-0.56	100	0.33	0.32	97	0.57	0.56	99	0.68	0.68	100	1.07	1.07	100	1.51	1.51	100
			BestSub	-0.56	-0.56	100	0.33	0.32	97	0.57	0.56	99	0.68	0.68	100	1.07	1.07	100	1.51	1.51	100

Table 17: **Scenario 4 (Imputation and Resampling) | Combination Method - Stability Selection | Imputation Method - Stochastic K-Nearest Neighbors Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.00	-0.00	15	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.13	0.04	34	0.49	0.49	100	0.99	0.98	100	1.49	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	0.13	0.04	34	0.49	0.49	100	0.99	0.98	100	1.49	1.49	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.02	-0.00	23	0.51	0.49	95	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.02	-0.00	23	0.51	0.49	95	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.07	0.05	73	0.50	0.49	99	0.98	0.97	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.00	-0.00	100	0.07	0.05	73	0.50	0.49	99	0.98	0.97	100	1.50	1.50	100	2.00	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	-0.01	-0.00	15	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.00	15	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	0.12	0.04	34	0.49	0.49	100	0.98	0.98	100	1.49	1.49	100	1.99	1.99	100
			BestSub	-0.01	-0.01	100	0.12	0.04	34	0.49	0.49	100	0.98	0.98	100	1.49	1.49	100	1.99	1.99	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.02	-0.00	22	0.51	0.48	94	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.02	-0.00	22	0.51	0.48	94	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.03	-0.03	100	0.08	0.05	68	0.50	0.49	99	0.98	0.98	100	1.49	1.49	100	1.99	1.98	100
			BestSub	-0.03	-0.03	100	0.08	0.05	68	0.50	0.49	99	0.98	0.98	100	1.49	1.49	100	1.99	1.98	100
MAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	0.00	0.00	20	0.51	0.50	100	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
			BestSub	0.00	0.00	100	0.00	0.00	20	0.51	0.50	100	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.00	-0.00	100	0.12	0.04	32	0.50	0.49	99	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	0.12	0.04	32	0.50	0.49	99	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
	High	**Ind**	**Forward**	-0.02	-0.02	100	-0.02	-0.00	19	0.53	0.45	86	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.02	-0.02	100	-0.02	-0.00	19	0.53	0.45	86	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.01	-0.01	100	0.05	0.04	65	0.52	0.46	89	1.00	0.99	99	1.50	1.50	100	2.01	2.01	100
			BestSub	-0.01	-0.01	100	0.05	0.04	65	0.52	0.46	89	1.00	0.99	99	1.50	1.50	100	2.01	2.01	100

Table 18: **Scenario 4 (Imputation and Resampling) | Combination Method - Stacking Selection | Imputation Method - Stochastic K-Nearest Neighbors Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.44	0.44		0.90	0.90		1.37	1.37		1.87	1.87	
			Ridge		0.00		-0.00	-0.00		0.42	0.42		0.85	0.85		1.30	1.30		1.76	1.76	
			Forward	0.00	0.00	100	-0.00	-0.00	90	0.44	0.44	100	0.91	0.90	100	1.38	1.38	100	1.88	1.87	100
			BestSub	0.00	0.00	100	-0.00	-0.00	90	0.44	0.44	100	0.91	0.90	100	1.38	1.38	100	1.88	1.87	100
		Corr	**LASSO**		0.00		0.12	0.12		0.52	0.52		0.96	0.96		1.42	1.42		1.90	1.90	
			Ridge		0.00		0.27	0.27		0.61	0.61		0.95	0.95		1.32	1.32		1.69	1.69	
			Forward	-0.00	-0.00	100	0.13	0.12	92	0.53	0.53	100	0.97	0.96	100	1.43	1.43	100	1.91	1.91	100
			BestSub	-0.00	-0.00	100	0.13	0.12	92	0.53	0.53	100	0.97	0.96	100	1.43	1.43	100	1.91	1.91	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.36	0.36		0.76	0.76		1.18	1.18		1.65	1.65	
			Ridge		0.00		-0.00	-0.00		0.35	0.35		0.73	0.72		1.13	1.12		1.56	1.56	
			Forward	0.00	0.00	100	-0.01	-0.01	91	0.37	0.37	100	0.77	0.76	100	1.19	1.19	100	1.66	1.65	100
			BestSub	0.00	0.00	100	-0.01	-0.01	91	0.37	0.37	100	0.77	0.76	100	1.19	1.19	100	1.66	1.65	100
		Corr	**LASSO**		0.00		0.29	0.29		0.60	0.60		0.92	0.92		1.31	1.31		1.72	1.72	
			Ridge		0.00		0.40	0.40		0.66	0.66		0.92	0.92		1.24	1.23		1.55	1.55	
			Forward	-0.00	-0.00	100	0.30	0.29	98	0.60	0.60	100	0.93	0.93	100	1.32	1.32	100	1.73	1.73	100
			BestSub	-0.00	-0.00	100	0.30	0.29	98	0.60	0.60	100	0.93	0.93	100	1.32	1.32	100	1.73	1.73	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.44	0.43		0.89	0.89		1.37	1.37		1.86	1.85	
			Ridge		0.00		-0.00	-0.00		0.42	0.42		0.85	0.85		1.29	1.29		1.75	1.75	
			Forward	0.08	0.08	100	-0.00	-0.00	88	0.44	0.44	100	0.90	0.90	100	1.37	1.37	100	1.86	1.86	100
			BestSub	0.08	0.08	100	-0.00	-0.00	88	0.44	0.44	100	0.90	0.90	100	1.37	1.37	100	1.86	1.86	100
		Corr	**LASSO**		0.00		0.12	0.12		0.53	0.53		0.96	0.96		1.42	1.42		1.90	1.90	
			Ridge		0.00		0.27	0.27		0.61	0.61		0.95	0.95		1.32	1.32		1.69	1.69	
			Forward	0.04	0.04	100	0.12	0.11	93	0.54	0.53	100	0.97	0.97	100	1.43	1.43	100	1.91	1.91	100
			BestSub	0.04	0.04	100	0.12	0.11	93	0.54	0.53	100	0.97	0.97	100	1.43	1.43	100	1.91	1.91	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.35	0.35		0.74	0.74		1.16	1.16		1.60	1.60	
			Ridge		0.00		-0.00	-0.00		0.34	0.34		0.71	0.71		1.10	1.10		1.52	1.52	
			Forward	0.07	0.07	100	-0.00	-0.00	89	0.36	0.36	99	0.75	0.75	100	1.16	1.16	100	1.61	1.61	100
			BestSub	0.07	0.07	100	-0.00	-0.00	89	0.36	0.36	99	0.75	0.75	100	1.16	1.16	100	1.61	1.61	100
		Corr	**LASSO**		0.00		0.29	0.29		0.59	0.59		0.93	0.93		1.31	1.31		1.71	1.70	
			Ridge		0.00		0.40	0.40		0.64	0.64		0.92	0.92		1.23	1.23		1.55	1.55	
			Forward	0.10	0.10	100	0.31	0.30	97	0.60	0.60	100	0.94	0.94	100	1.31	1.31	100	1.71	1.71	100
			BestSub	0.10	0.10	100	0.31	0.30	97	0.60	0.60	100	0.94	0.94	100	1.31	1.31	100	1.71	1.71	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.41	0.40		0.93	0.93		1.39	1.39		1.87	1.86	
			Ridge		0.00		-0.00	-0.00		0.39	0.39		0.88	0.87		1.31	1.31		1.76	1.75	
			Forward	0.02	0.02	100	-0.00	-0.00	90	0.41	0.41	100	0.93	0.93	100	1.39	1.39	100	1.87	1.87	100
			BestSub	0.02	0.02	100	-0.00	-0.00	90	0.41	0.41	100	0.93	0.93	100	1.39	1.39	100	1.87	1.87	100
		Corr	**LASSO**		0.00		0.11	0.11		0.46	0.46		0.96	0.96		1.43	1.43		1.88	1.88	
			Ridge		0.00		0.26	0.26		0.56	0.56		0.95	0.95		1.32	1.32		1.67	1.67	
			Forward	-0.07	-0.07	100	0.11	0.10	92	0.47	0.47	100	0.97	0.97	100	1.44	1.44	100	1.89	1.89	100
			BestSub	-0.07	-0.07	100	0.11	0.10	92	0.47	0.47	100	0.97	0.97	100	1.44	1.44	100	1.89	1.89	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.27	0.27		0.79	0.79		1.20	1.20		1.68	1.68	
			Ridge		0.00		-0.00	-0.00		0.26	0.26		0.75	0.75		1.13	1.13		1.59	1.58	
			Forward	-0.04	-0.04	100	-0.00	-0.00	89	0.28	0.28	100	0.80	0.80	100	1.20	1.20	100	1.69	1.69	100
			BestSub	-0.04	-0.04	100	-0.00	-0.00	89	0.28	0.28	100	0.80	0.80	100	1.20	1.20	100	1.69	1.69	100
		Corr	**LASSO**		0.00		0.25	0.25		0.44	0.44		0.71	0.71		1.10	1.10		1.56	1.56	
			Ridge		0.00		0.33	0.33		0.50	0.50		0.74	0.74		1.05	1.05		1.41	1.41	
			Forward	-0.58	-0.58	100	0.27	0.26	96	0.45	0.45	99	0.72	0.71	100	1.11	1.11	100	1.56	1.56	100
			BestSub	-0.58	-0.58	100	0.27	0.26	96	0.45	0.45	99	0.72	0.71	100	1.11	1.11	100	1.56	1.56	100

Table 19: **Scenario 4 (Imputation and Resampling) | Combination Method - Stability Selection | Imputation Method - Deterministic Regression Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.01	-0.00	51	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	-0.04	-0.02	47	0.51	0.50	99	1.00	1.00	100	1.51	1.50	100	2.01	2.01	100
			BestSub	0.00	0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.01	0.01	98	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.00	2.00	100
			BestSub	-0.00	-0.00	100	0.01	0.01	100	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	-0.01	-0.00	48	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	-0.04	-0.02	44	0.51	0.50	99	1.00	1.00	100	1.51	1.50	100	2.01	2.01	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.04	-0.04	100	0.00	0.00	93	0.51	0.50	99	0.99	0.99	100	1.50	1.50	100	1.99	1.99	100
			BestSub	-0.04	-0.04	100	0.01	0.01	100	0.50	0.50	100	0.99	0.98	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**Forward**	0.01	0.01	100	-0.01	-0.00	49	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	-0.01	-0.01	46	0.51	0.51	100	1.00	1.00	100	1.51	1.51	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.01	-0.01	100	-0.01	-0.01	96	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.01	-0.01	100	-0.00	-0.00	91	0.50	0.50	99	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100
			BestSub	-0.00	-0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100

Table 20: **Scenario 4 (Imputation and Resampling) | Combination Method - Stacking Selection | Imputation Method - Deterministic Regression Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.52	0.52		1.03	1.03		1.53	1.53		2.02	2.02	
			Ridge		0.00		-0.00	-0.00		0.50	0.50		0.97	0.97		1.44	1.44		1.90	1.90	
			Forward	0.00	0.00	100	-0.00	-0.00	92	0.53	0.53	100	1.04	1.04	100	1.54	1.54	100	2.03	2.03	100
			BestSub	0.00	0.00	100	-0.00	-0.00	92	0.53	0.53	100	1.04	1.04	100	1.54	1.54	100	2.03	2.03	100
		Corr	**LASSO**		0.00		-0.01	-0.01		0.48	0.48		1.00	1.00		1.52	1.52		2.02	2.02	
			Ridge		0.00		0.17	0.17		0.59	0.59		0.99	0.99		1.39	1.39		1.78	1.78	
			Forward	0.00	0.00	100	-0.05	-0.05	93	0.49	0.49	100	1.02	1.01	100	1.53	1.53	100	2.04	2.03	100
			BestSub	0.00	0.00	100	-0.05	-0.05	93	0.49	0.49	100	1.02	1.01	100	1.53	1.53	100	2.04	2.03	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.53	0.53		1.05	1.05		1.55	1.55		2.04	2.03	
			Ridge		0.00		-0.01	-0.01		0.51	0.51		0.99	0.99		1.46	1.45		1.91	1.90	
			Forward	-0.00	-0.00	100	-0.01	-0.01	96	0.54	0.54	100	1.06	1.06	100	1.56	1.56	100	2.04	2.04	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	96	0.54	0.54	100	1.06	1.06	100	1.56	1.56	100	2.04	2.04	100
		Corr	**LASSO**		0.00		-0.03	-0.03		0.48	0.48		1.00	1.00		1.54	1.54		2.04	2.04	
			Ridge		0.00		0.18	0.18		0.60	0.60		0.99	0.99		1.39	1.39		1.77	1.77	
			Forward	-0.00	-0.00	100	-0.07	-0.07	96	0.49	0.49	100	1.02	1.02	100	1.55	1.55	100	2.05	2.05	100
			BestSub	-0.00	-0.00	100	-0.07	-0.07	96	0.49	0.49	100	1.02	1.02	100	1.55	1.55	100	2.05	2.05	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.52	0.52		1.03	1.03		1.53	1.52		2.02	2.02	
			Ridge		0.00		-0.01	-0.01		0.50	0.50		0.97	0.97		1.43	1.43		1.89	1.89	
			Forward	0.08	0.08	100	-0.01	-0.01	94	0.53	0.53	100	1.04	1.04	100	1.53	1.53	100	2.03	2.02	100
			BestSub	0.08	0.08	100	-0.01	-0.01	94	0.53	0.53	100	1.04	1.04	100	1.53	1.53	100	2.03	2.02	100
		Corr	**LASSO**		0.00		-0.01	-0.01		0.48	0.48		1.00	1.00		1.52	1.52		2.02	2.02	
			Ridge		0.00		0.17	0.17		0.59	0.59		0.99	0.99		1.39	1.39		1.78	1.77	
			Forward	0.05	0.05	100	-0.05	-0.05	93	0.50	0.50	100	1.02	1.02	100	1.53	1.53	100	2.03	2.03	100
			BestSub	0.05	0.05	100	-0.05	-0.05	93	0.50	0.50	100	1.02	1.02	100	1.53	1.53	100	2.03	2.03	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.53	0.53		1.05	1.04		1.54	1.54		2.03	2.02	
			Ridge		0.00		-0.00	-0.00		0.50	0.50		0.98	0.98		1.44	1.44		1.90	1.90	
			Forward	0.22	0.22	100	-0.01	-0.01	96	0.54	0.54	100	1.05	1.05	100	1.55	1.55	100	2.04	2.03	100
			BestSub	0.22	0.22	100	-0.01	-0.01	96	0.54	0.54	100	1.05	1.05	100	1.55	1.55	100	2.04	2.03	100
		Corr	**LASSO**		0.00		-0.01	-0.01		0.48	0.48		1.01	1.01		1.53	1.53		2.03	2.03	
			Ridge		0.00		0.19	0.19		0.60	0.60		1.00	1.00		1.39	1.39		1.77	1.77	
			Forward	0.15	0.15	100	-0.05	-0.05	96	0.50	0.50	99	1.03	1.02	100	1.55	1.55	100	2.05	2.05	100
			BestSub	0.15	0.15	100	-0.05	-0.05	96	0.50	0.50	99	1.03	1.02	100	1.55	1.55	100	2.05	2.05	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.56	0.56		0.99	0.99		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.00	-0.00		0.53	0.53		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	0.03	0.03	100	-0.00	-0.00	94	0.57	0.57	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.03	0.03	100	-0.00	-0.00	94	0.57	0.57	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.02	0.02		0.56	0.56		0.97	0.97		1.48	1.48		1.97	1.97	
			Ridge		0.00		0.21	0.21		0.66	0.66		0.95	0.95		1.35	1.35		1.74	1.74	
			Forward	0.01	0.01	100	-0.02	-0.02	94	0.57	0.57	100	0.98	0.98	100	1.49	1.49	100	1.99	1.98	100
			BestSub	0.01	0.01	100	-0.02	-0.02	94	0.57	0.57	100	0.98	0.98	100	1.49	1.49	100	1.99	1.98	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.55	0.55		0.99	0.99		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.01	-0.01		0.52	0.52		0.93	0.93		1.40	1.40		1.86	1.86	
			Forward	0.06	0.06	100	-0.01	-0.01	95	0.56	0.56	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.06	0.06	100	-0.01	-0.01	95	0.56	0.56	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.03	0.03		0.55	0.55		0.96	0.96		1.47	1.47		1.98	1.97	
			Ridge		0.00		0.30	0.30		0.70	0.70		0.92	0.92		1.33	1.32		1.71	1.71	
			Forward	0.03	0.03	100	-0.01	-0.01	94	0.57	0.57	100	0.98	0.97	100	1.49	1.49	100	1.99	1.99	100
			BestSub	0.03	0.03	100	-0.01	-0.01	94	0.57	0.57	100	0.98	0.97	100	1.49	1.49	100	1.99	1.99	100

Table 21: **Scenario 4 (Imputation and Resampling) | Combination Method - Stability Selection | Imputation Method - Stochastic Regression Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**Forward**	0.00	0.00	100	-0.01	-0.00	28	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.05	0.02	29	0.50	0.49	99	0.99	0.99	100	1.50	1.50	100	2.01	2.00	100
			BestSub	0.00	0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.01	2.01	100
	High	**Ind**	**Forward**	-0.01	-0.01	100	-0.03	-0.01	39	0.51	0.50	99	1.00	1.00	100	1.51	1.51	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.00	-0.00	100	0.12	0.07	57	0.50	0.49	98	0.98	0.98	100	1.49	1.49	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	0.01	0.01	100	0.51	0.51	100	0.98	0.98	100	1.51	1.51	100	2.01	2.00	100
MNAR	**Low**	**Ind**	**Forward**	-0.02	-0.02	100	-0.00	-0.00	28	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	2.00	1.99	100
			BestSub	-0.02	-0.02	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.50	1.50	100	1.99	1.99	100
		Corr	**Forward**	-0.01	-0.01	100	0.04	0.01	30	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.00	-0.00	100	0.50	0.50	100	0.99	0.99	100	1.50	1.50	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.07	-0.07	100	-0.02	-0.01	31	0.50	0.50	99	0.99	0.99	100	1.49	1.49	100	1.98	1.98	100
			BestSub	-0.07	-0.07	100	-0.01	-0.01	100	0.50	0.50	100	1.00	0.99	100	1.49	1.49	100	1.98	1.98	100
		Corr	**Forward**	-0.03	-0.03	100	0.15	0.07	50	0.50	0.49	99	0.97	0.97	100	1.49	1.48	100	1.98	1.98	100
			BestSub	-0.04	-0.04	100	0.01	0.01	100	0.50	0.50	100	0.99	0.98	100	1.50	1.50	100	1.99	1.99	100
MAR	**Low**	**Ind**	**Forward**	0.01	0.01	100	-0.01	-0.00	32	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.51	0.51	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**Forward**	0.00	0.00	100	0.02	0.01	31	0.50	0.50	99	1.00	0.99	100	1.51	1.50	100	2.00	2.00	100
			BestSub	0.00	0.00	100	-0.00	-0.00	100	0.50	0.50	100	1.00	0.99	100	1.51	1.51	100	2.00	2.00	100
	High	**Ind**	**Forward**	-0.00	-0.00	100	-0.02	-0.01	29	0.51	0.50	98	1.00	1.00	100	1.50	1.49	100	2.00	2.00	100
			BestSub	-0.01	-0.01	100	-0.01	-0.01	100	0.50	0.50	100	1.00	1.00	100	1.49	1.49	100	2.00	2.00	100
		Corr	**Forward**	-0.00	-0.00	100	0.09	0.02	23	0.53	0.48	92	0.99	0.99	100	1.51	1.51	100	2.01	2.00	100
			BestSub	-0.00	-0.00	100	0.00	0.00	100	0.50	0.50	100	0.99	0.99	100	1.51	1.50	100	2.01	2.00	100

Table 22: **Scenario 4 (Imputation and Resampling) | Combination Method - Stacking Selection | Imputation Method - Stochastic Regression Imputation**

Miss Mech	Miss Rate	Cor	Sel	$\hat{\beta}_0$ NA	$\hat{\beta}_0$ 0	$\hat{\beta}_0$ %	$\hat{\beta}_1$ NA	$\hat{\beta}_1$ 0	$\hat{\beta}_1$ %	$\hat{\beta}_2$ NA	$\hat{\beta}_2$ 0	$\hat{\beta}_2$ %	$\hat{\beta}_3$ NA	$\hat{\beta}_3$ 0	$\hat{\beta}_3$ %	$\hat{\beta}_4$ NA	$\hat{\beta}_4$ 0	$\hat{\beta}_4$ %	$\hat{\beta}_5$ NA	$\hat{\beta}_5$ 0	$\hat{\beta}_5$ %
MCAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.48	0.48		0.97	0.97		1.46	1.46		1.96	1.96	
			Ridge		0.00		-0.00	-0.00		0.46	0.46		0.92	0.92		1.38	1.37		1.84	1.84	
			Forward	-0.00	-0.00	100	-0.00	-0.00	90	0.49	0.49	100	0.98	0.98	100	1.47	1.47	100	1.97	1.97	100
			BestSub	-0.00	-0.00	100	-0.00	-0.00	90	0.49	0.49	100	0.98	0.98	100	1.47	1.47	100	1.97	1.97	100
		Corr	**LASSO**		0.00		0.04	0.04		0.50	0.50		0.97	0.97		1.47	1.47		1.98	1.98	
			Ridge		0.00		0.21	0.21		0.59	0.59		0.97	0.96		1.36	1.35		1.76	1.75	
			Forward	-0.00	-0.00	100	0.03	0.02	90	0.51	0.51	100	0.99	0.99	100	1.48	1.48	100	1.99	1.99	100
			BestSub	-0.00	-0.00	100	0.03	0.02	90	0.51	0.51	100	0.99	0.99	100	1.48	1.48	100	1.99	1.99	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.43	0.43		0.85	0.85		1.35	1.34		1.81	1.81	
			Ridge		0.00		-0.01	-0.01		0.41	0.41		0.80	0.80		1.27	1.27		1.70	1.70	
			Forward	0.04	0.04	100	-0.01	-0.01	90	0.44	0.44	100	0.85	0.85	100	1.35	1.35	100	1.81	1.81	100
			BestSub	0.04	0.04	100	-0.01	-0.01	90	0.44	0.44	100	0.85	0.85	100	1.35	1.35	100	1.81	1.81	100
		Corr	**LASSO**		0.00		0.14	0.14		0.55	0.55		0.92	0.92		1.43	1.43		1.89	1.88	
			Ridge		0.00		0.28	0.28		0.63	0.63		0.92	0.92		1.33	1.32		1.68	1.68	
			Forward	0.03	0.03	100	0.14	0.13	93	0.56	0.56	100	0.93	0.92	100	1.44	1.44	100	1.89	1.89	100
			BestSub	0.03	0.03	100	0.14	0.13	93	0.56	0.56	100	0.93	0.92	100	1.44	1.44	100	1.89	1.89	100
MNAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.48	0.48		0.97	0.97		1.46	1.45		1.95	1.95	
			Ridge		0.00		-0.00	-0.00		0.46	0.46		0.92	0.91		1.37	1.37		1.83	1.83	
			Forward	0.06	0.06	100	-0.01	-0.01	90	0.49	0.49	100	0.98	0.98	100	1.46	1.46	100	1.96	1.96	100
			BestSub	0.06	0.06	100	-0.01	-0.01	90	0.49	0.49	100	0.98	0.98	100	1.46	1.46	100	1.96	1.96	100
		Corr	**LASSO**		0.00		0.04	0.04		0.50	0.50		0.98	0.98		1.47	1.47		1.97	1.97	
			Ridge		0.00		0.21	0.21		0.59	0.59		0.97	0.97		1.35	1.35		1.75	1.75	
			Forward	0.05	0.05	100	0.03	0.02	92	0.51	0.51	100	0.99	0.99	100	1.48	1.48	100	1.99	1.98	100
			BestSub	0.05	0.05	100	0.03	0.02	92	0.51	0.51	100	0.99	0.99	100	1.48	1.48	100	1.99	1.98	100
	High	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.43	0.43		0.84	0.83		1.33	1.33		1.78	1.78	
			Ridge		0.00		-0.00	-0.00		0.41	0.41		0.79	0.79		1.25	1.25		1.68	1.68	
			Forward	0.17	0.17	100	-0.00	-0.00	90	0.44	0.44	100	0.84	0.84	100	1.33	1.33	100	1.78	1.78	100
			BestSub	0.17	0.17	100	-0.00	-0.00	90	0.44	0.44	100	0.84	0.84	100	1.33	1.33	100	1.78	1.78	100
		Corr	**LASSO**		0.00		0.17	0.17		0.58	0.58		0.93	0.93		1.42	1.42		1.86	1.86	
			Ridge		0.00		0.30	0.30		0.65	0.65		0.92	0.92		1.32	1.32		1.67	1.67	
			Forward	0.18	0.18	100	0.17	0.16	95	0.59	0.58	100	0.93	0.93	100	1.43	1.43	100	1.87	1.87	100
			BestSub	0.18	0.18	100	0.17	0.16	95	0.59	0.58	100	0.93	0.93	100	1.43	1.43	100	1.87	1.87	100
MAR	**Low**	**Ind**	**LASSO**		0.00		-0.00	-0.00		0.47	0.47		0.99	0.99		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.00	-0.00		0.44	0.44		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	0.03	0.03	100	-0.00	-0.00	92	0.48	0.47	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	0.03	0.03	100	-0.00	-0.00	92	0.48	0.47	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.03	0.03		0.46	0.46		0.99	0.99		1.50	1.50		2.00	1.99	
			Ridge		0.00		0.20	0.20		0.56	0.56		0.98	0.98		1.38	1.38		1.77	1.77	
			Forward	0.02	0.02	100	0.01	0.01	91	0.47	0.47	100	1.01	1.00	100	1.52	1.51	100	2.01	2.01	100
			BestSub	0.02	0.02	100	0.01	0.01	91	0.47	0.47	100	1.01	1.00	100	1.52	1.51	100	2.01	2.01	100
	High	**Ind**	**LASSO**		0.00		-0.01	-0.01		0.31	0.31		0.99	0.99		1.49	1.49		1.99	1.99	
			Ridge		0.00		-0.01	-0.01		0.29	0.29		0.93	0.93		1.40	1.40		1.87	1.87	
			Forward	-0.03	-0.03	100	-0.01	-0.01	90	0.31	0.31	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
			BestSub	-0.03	-0.03	100	-0.01	-0.01	90	0.31	0.31	100	1.00	1.00	100	1.50	1.50	100	2.00	2.00	100
		Corr	**LASSO**		0.00		0.04	0.04		0.29	0.29		1.02	1.01		1.53	1.53		2.04	2.04	
			Ridge		0.00		0.20	0.20		0.42	0.42		1.01	1.01		1.42	1.41		1.81	1.81	
			Forward	-0.02	-0.02	100	0.03	0.03	87	0.30	0.30	100	1.03	1.02	100	1.54	1.54	100	2.05	2.05	100
			BestSub	-0.02	-0.02	100	0.03	0.03	87	0.30	0.30	100	1.03	1.02	100	1.54	1.54	100	2.05	2.05	100